

AWS Lab 02: Theory

- Data Centre
- Auto Scaling
 - Scale-up
 - Scale-down
- Cloud -> Make things scalable according to the usage in real-time and robust
- Horizontal scaling and vertical scaling
- Load Balancer: Maximum concurrent users or asking for something/lag in DB "Slowing machine because of any reason" so we balance load
- Docker -> Containerization
- Use case of Docker: Making the environment portable.
- Lab Today: Setting up load balancer and how it works

Elastic Load Balancing (ELB)

Dividing load to different machine

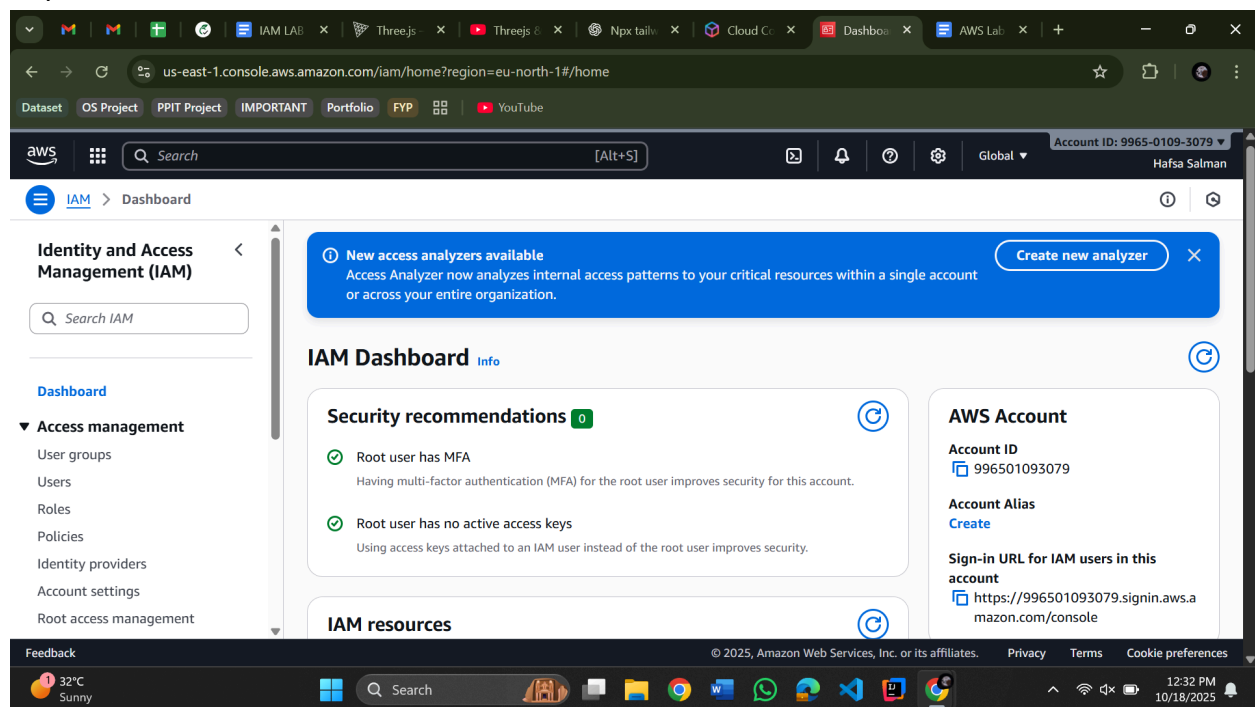
Security group -> Define inbounds and outbounds

Target group -> Set load balancer to define which machine to give/take services

//User Account

- Login using root account
- Open incognito
- Login through IAM by entering Account ID (copy it from root account), user name (the one you made in root account) and user password
- Follow the following steps

//Open IAM



//Open EC2

The screenshot displays the AWS Management Console for the EC2 service in the Europe (Stockholm) region. The interface includes a top navigation bar with the AWS logo, a search bar, and the account ID 9965-0109-3079. The left sidebar shows the EC2 dashboard and a list of instances. The main content area features a 'Resources' section with a table of EC2 resources and their counts. To the right, there are sections for 'Account attributes' and 'Settings'. The bottom of the console shows a 'Launch instance' button and a 'Service health' section. The Windows taskbar at the bottom indicates the system time is 12:33 PM on 10/18/2025.

Resources	
Instances (running)	0
Capacity Reservations	0
Elastic IPs	0
Key pairs	2
Placement groups	0
Snapshots	0
Auto Scaling Groups	0
Dedicated Hosts	0
Instances	0
Load balancers	0
Security groups	3
Volumes	0

//create Security groups (top right)

The image shows two screenshots of the AWS Management Console. The top screenshot displays the 'Security Groups' page for the 'eu-north-1' region. It lists three security groups: 'my-security-group' (ID: sg-0fd2a2dbc51b553db), 'HafsaClassSecurity' (ID: sg-038f27f1cb0253be6), and 'default' (ID: sg-0f93fe7f396203778). The bottom screenshot shows the 'Create security group' wizard. In the 'Basic details' section, the 'Security group name' is 'Security-group-for-ec2', the 'Description' is 'Security-group-for-ec2', and the 'VPC' is 'vpc-08f7b340e0705d05b'. The 'Inbound rules' section is partially visible.

Security Groups (3)

Name	Security group ID	Security group name	VPC ID
-	sg-0fd2a2dbc51b553db	my-security-group	vpc-08f7b340e0705d05b
-	sg-038f27f1cb0253be6	HafsaClassSecurity	vpc-08f7b340e0705d05b
-	sg-0f93fe7f396203778	default	vpc-08f7b340e0705d05b

Create security group

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

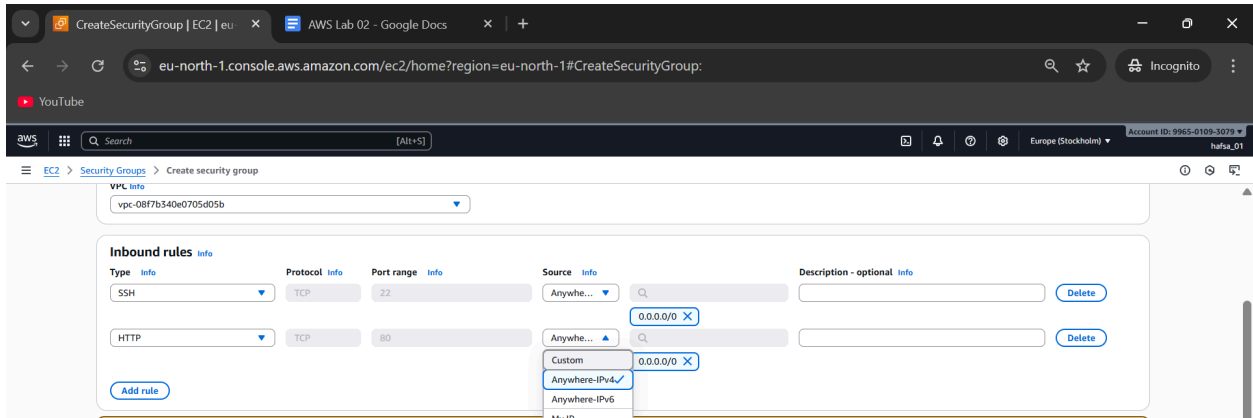
Security group name Info
Security-group-for-ec2
Name cannot be edited after creation.

Description Info
Security-group-for-ec2

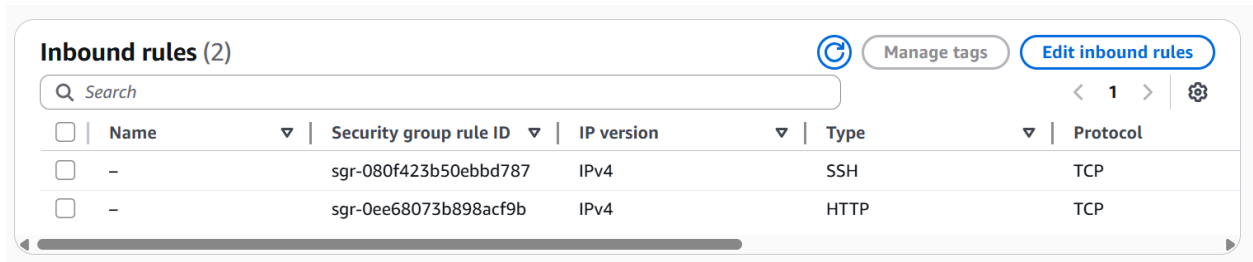
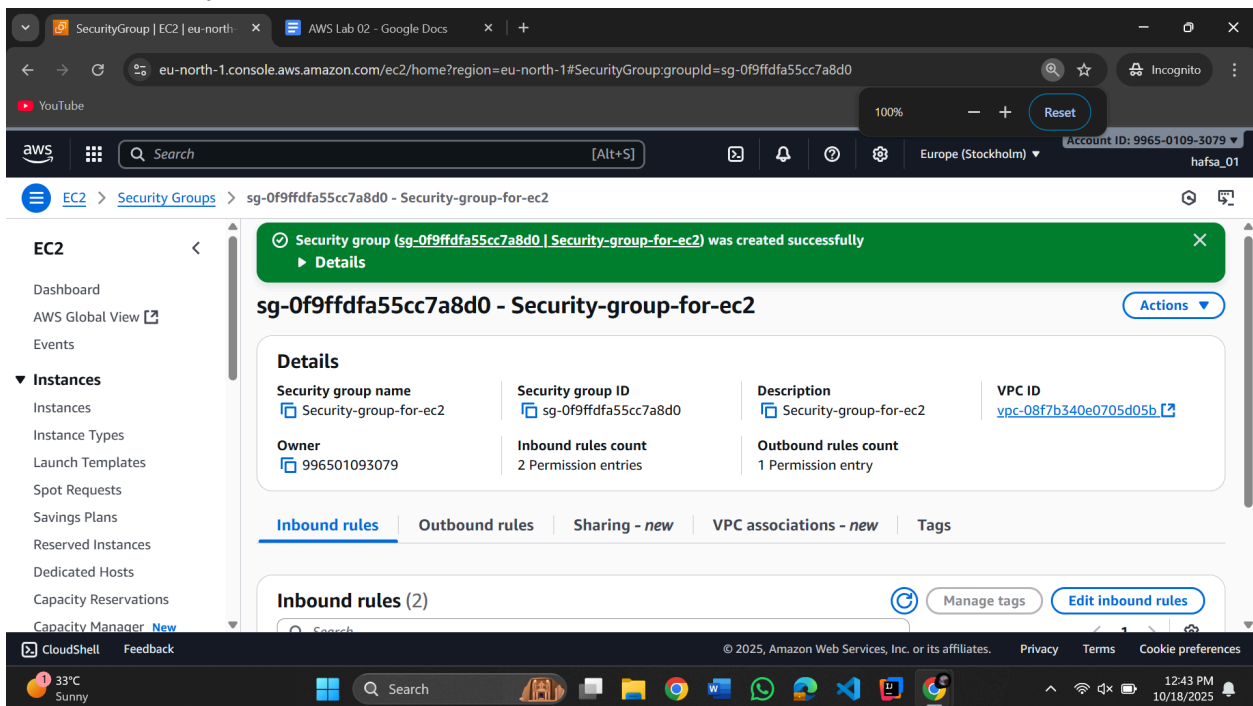
VPC Info
vpc-08f7b340e0705d05b

Inbound rules Info

//Source: Anywhere Ipv4



//Create Security Group



//Create another Security group

EC2 > Security Groups > sg-0f9ffdfa55cc7a8d0 - Security-group-for-ec2

Images

AMI Catalog

Elastic Block Store

Network & Security

Load Balancing

Security group name

Security group ID

Description

VPC ID

Owner

Inbound rules count

Outbound rules count

Security-group-for-ec2

sg-0f9ffdfa55cc7a8d0

Security-group-for-ec2

vpc-08f7b340e0705d05b

996501093079

2 Permission entries

1 Permission entry

Inbound rules

Outbound rules

Sharing - new

VPC associations - new

Tags

Inbound rules (2)

Search

Manage tags

Edit inbound rules

	Name	Security group rule ID	IP version	Type	Protocol
☐	-	sgr-080f423b50ebbd787	IPv4	SSH	TCP
☐	-	sgr-0ee68073b898acf9b	IPv4	HTTP	TCP

CloudShell

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CreateSecurityGroup | EC2 | eu

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eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#CreateSecurityGroup:

aws

Search

[Alt+S]

Europe (Stockholm)

Account ID: 9965-0109-3079

hafsa_01

EC2 > Security Groups > Create security group

Create security group

Create security group

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name

Description

VPC

Inbound rules

CloudShell

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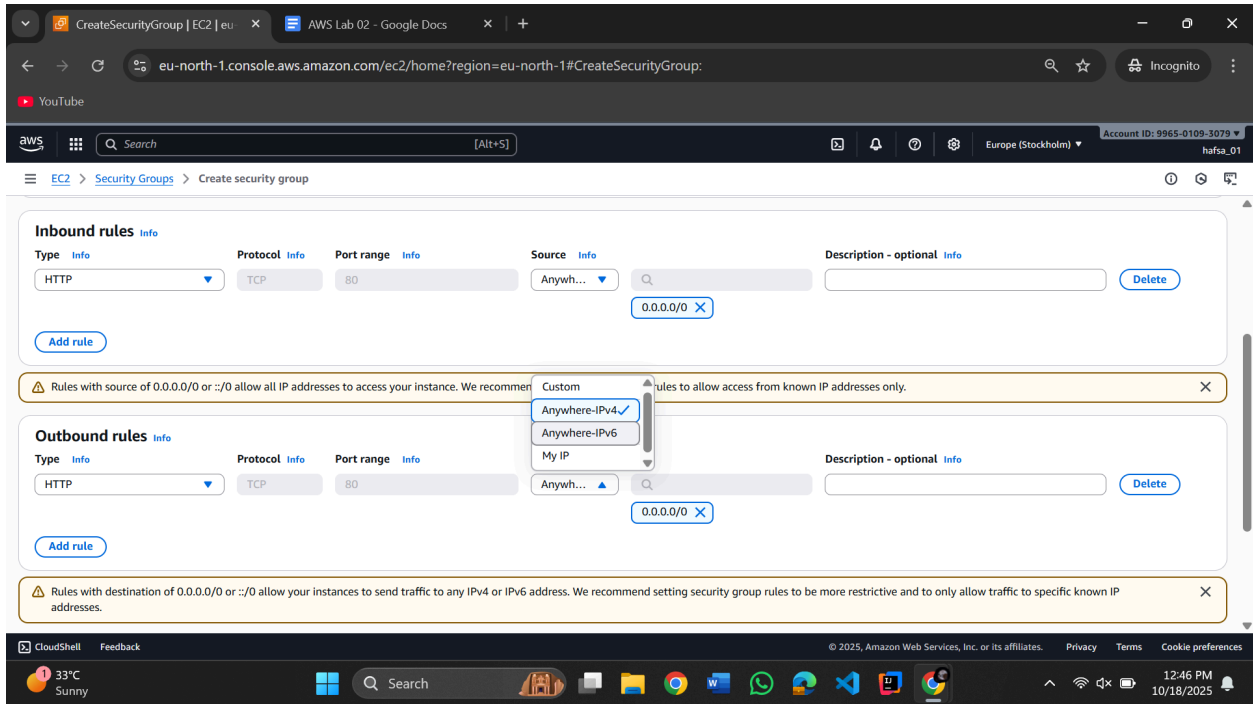
Terms

Cookie preferences

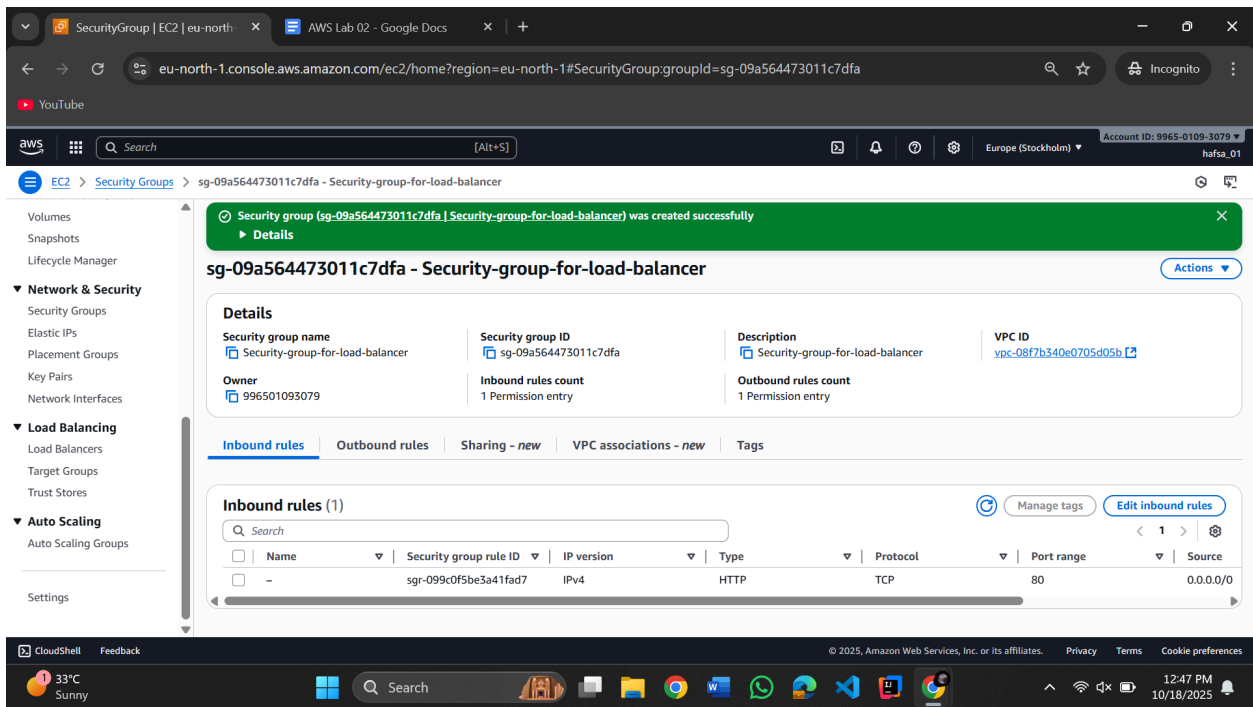
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12:45 PM 10/18/2025



(Source: Anywhere-IPv4)



//Make a target group (that has 2 EC2)

//Making EC2

Dashboard | EC2 | eu-north-1 x AWS Lab 02 - Google Docs x +

eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#Home:

YouTube

aws Search [Alt+S] Europe (Stockholm) Account ID: 9965-0109-3079 hafsa_01

EC2

[Dashboard](#)

[AWS Global View](#)

[Events](#)

Instances

[Instances](#)

[Instance Types](#)

[Launch Templates](#)

[Spot Requests](#)

[Savings Plans](#)

[Reserved Instances](#)

[Dedicated Hosts](#)

[Capacity Reservations](#)

[Capacity Manager](#) [New](#)

Images

[AMIs](#)

[AMI Catalog](#)

Elastic Block Store

Resources

You are using the following Amazon EC2 resources in the Europe (Stockholm) Region:

Instances (running)	0	Auto Scaling Groups	0	Capacity Reservations	0
Dedicated Hosts	0	Elastic IPs	0	Instances	0
Key pairs	2	Load balancers	0	Placement groups	0
Security groups	3	Snapshots	0	Volumes	0

Launch instance

To get started, launch an Amazon EC2 instance, which is a virtual server in the cloud.

[Launch instance](#) [Migrate a server](#)

Note: Your instances will launch in the Europe (Stockholm) Region

Service health

[AWS Health Dashboard](#)

An error occurred
An error occurred retrieving service health information

[Diagnose with Amazon Q](#)

EC2 Free Tier

[Info](#)
Offers for all AWS Regions.

0 EC2 free tier offers in use

End of month forecast

⚠ User: arn:aws:iam::996501093079:user/hafsa_01 is not authorized to perform: freetier:GetFreeTierUsage on resource: arn:aws:freetier:us-east-1:996501093079:/GetFreeTierUsage because no identity-based policy allows the freetier:GetFreeTierUsage action

Exceeds free tier

⚠ User: arn:aws:iam::996501093079:user/hafsa_01 is not authorized to perform: freetier:GetFreeTierUsage on resource: arn:aws:freetier:us-east-1:996501093079:/GetFreeTierUsage because no identity-based policy allows the freetier:GetFreeTierUsage action

[View Global EC2 resources](#)

[View all AWS Free Tier offers](#)

https://eu-north-1.console.aws.amazon.com/console/home?region=eu-north-1

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(Launch Instance)

Launch an instance | EC2 | eu-north-1 x AWS Lab 02 - Google Docs x +

eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#LaunchInstances:

YouTube

aws Search [Alt+S] Europe (Stockholm) Account ID: 9965-0109-3079 hafsa_01

[EC2](#) > [Instances](#) > Launch an instance

Name and tags

[Info](#)

Name

[Add additional tags](#)

Application and OS Images (Amazon Machine Image)

[Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

[Recents](#) [Quick Start](#)



Amazon Linux



macOS



Ubuntu



Windows



Red Hat



SUSE Linux



Debian

[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

Summary

[Number of instances](#) [Info](#)

[Software Image \(AMI\)](#)
Canonical, Ubuntu, 24.04, amd64...[read more](#)
ami-0a716d3f3b16d290c

[Virtual server type \(instance type\)](#)
t3.micro

[Firewall \(security group\)](#)
New security group

[Storage \(volumes\)](#)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

Amazon Machine Image (AMI)

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Launch an instance | EC2 | eu-north-1 | AWS Lab 02 - Google Docs

eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#LaunchInstances:

Key pair (login) info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

Select

Create new key pair

Network settings info

Network

vpc-08f7b340e0705d05b

Subnet

No preference (Default subnet in any availability zone)

Auto-assign public IP

Enable

Firewall (security groups) info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group

Select existing security group

We'll create a new security group called 'launch-wizard-1' with the following rules:

Summary

Number of instances

1

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd64...read more

ami-0a716d3f3b16d290c

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

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Search

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Launch an instance | EC2 | eu-north-1 | AWS Lab 02 - Google Docs

eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#LaunchInstances:

Key pair (login) info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

Select

Network settings info

Network

vpc-08f7b340e0705d05b

Subnet

No preference (Default subnet in any availability zone)

Auto-assign public IP

Enable

Firewall (security groups) info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group

Select existing security group

We'll create a new security group called 'launch-wizard-1' with the following rules:

Create key pair

Key pair name

server-1-key

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

RSA

ED25519

Private key file format

.pem

.ppk

When prompted, store the private key in a secure and accessible location on your computer. You will need it later to connect to your instance. Learn more

Cancel

Create key pair

Summary

Number of instances

1

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd64...read more

ami-0a716d3f3b16d290c

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

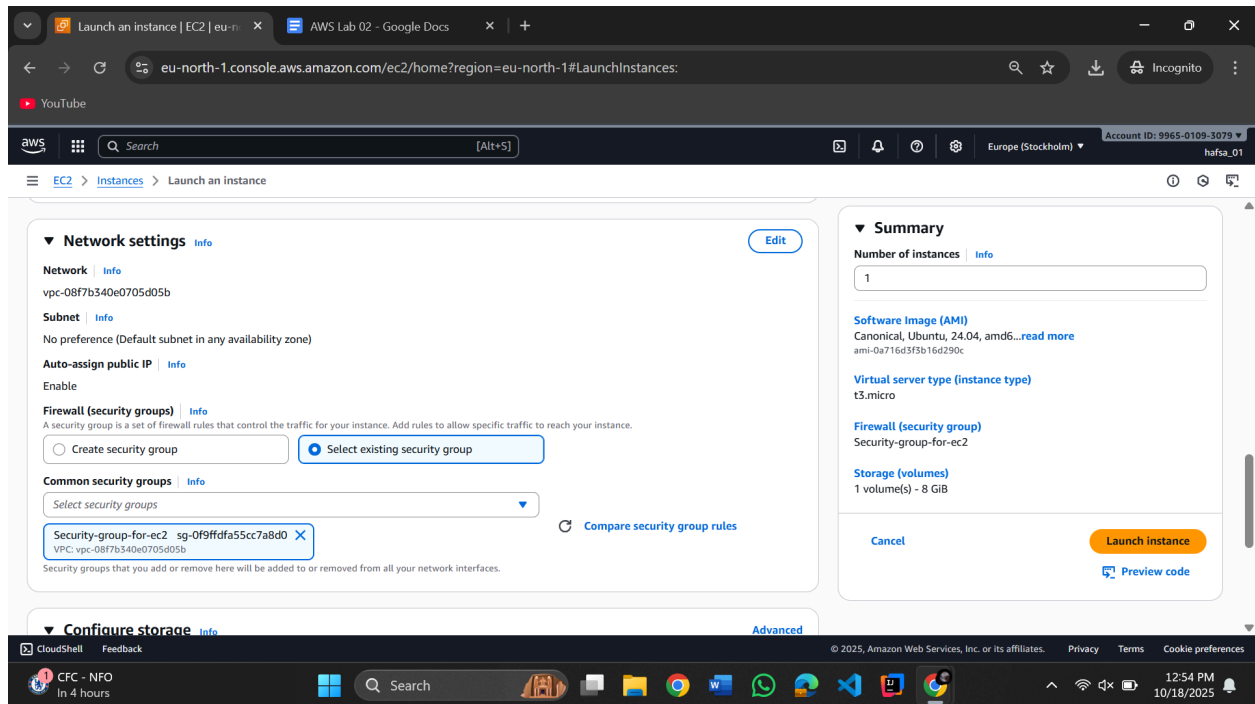
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Search

12:50 PM 10/18/2025



```
//Code
#!/bin/bash
yes | sudo apt update
yes | sudo apt install apache2 -y

HOSTNAME=$(hostname)
IP_ADDRESS=$(hostname -I | awk '{print $1}')
echo "<h1>IP Address of this server is</h1><p><strong>$IP_ADDRESS</strong></p>" | sudo
tee /var/www/html/index.html
sudo systemctl restart apache2

//In advance detail -> Copy paste this above code
```

The image displays two sequential screenshots of the AWS Management Console, specifically the EC2 'Launch instance' page, illustrating the process of creating a new EC2 instance.

Top Screenshot: Launch instance page

- Browser:** The browser address bar shows `eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#LaunchInstances:`.
- Page Header:** The header indicates the user is logged in as `hafsai_01` with account ID `9965-0109-3079`.
- Navigation:** The left sidebar shows the navigation menu with `EC2 > Instances > Launch an instance`.
- User data (optional):** A text area contains the following script:

```
#!/bin/bash
yes | sudo apt update
yes | sudo apt install apache2 -y

HOSTNAME=$(hostname)
IP_ADDRESS=$(hostname -I | awk '{print $1}')
echo "<h1>IP Address of this server is</h1><p><strong>$IP_ADDRESS</strong></p>" | sudo tee /var/www/html/index.html
sudo systemctl restart apache2
```
- Summary:** The right sidebar shows the configuration summary:
 - Number of instances:** 1
 - Software Image (AMI):** Canonical, Ubuntu, 24.04, amd64...read more
 - Virtual server type (instance type):** t3.micro
 - Firewall (security group):** Security-group-for-ec2
 - Storage (volumes):** 1 volume(s) - 8 GiB
- Buttons:** `Cancel` and `Launch instance` buttons are visible.

Bottom Screenshot: Success and Next Steps

- Success:** A green banner at the top states: `Successfully initiated launch of instance (i-0fd8ba8cac3f59b40)`.
- Launch log:** A section for viewing the launch log.
- Next Steps:** A section with a search bar and a list of recommended actions:
 - Create billing usage alerts:** To manage costs and avoid surprise bills, set up email notifications for billing usage thresholds. `Create billing alerts` button.
 - Connect to your instance:** Once your instance is running, log into it from your local computer. `Connect to instance` button.
 - Connect an RDS database:** Configure the connection between an EC2 instance and a database to allow traffic flow between them. `Connect an RDS database` button.
 - Create EBS snapshot policy:** Create a policy that automates the creation, retention, and deletion of EBS snapshots. `Create EBS snapshot policy` button.

//Making Second EC2

Launch an instance | EC2 | eu-north-1

AWS Lab 02 - Google Docs

AirForShare.com - Virtual Desk

+

eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#LaunchInstances:

YouTube

aws

Search

[Alt+S]

Europe (Stockholm)

Account ID: 9965-0109-3079

hafsa_01

EC2 > Instances > Launch an instance

It seems like you may be new to launching Instances in EC2. Take a walkthrough to learn about EC2, how to launch instances and about best practices.

Take a walkthrough

Do not show me this message again.

Launch an instance

Info

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

Info

Name

server-4

Add additional tags

Application and OS Images (Amazon Machine Image)

Info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

Recents

Quick Start

Summary

Number of instances

Info

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.9.2...read more

ami-0854d4f8e4bd6b834

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

CloudShell

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Sunny

Search

1:00 PM

10/18/2025

Launch an instance | EC2 | eu-north-1

AWS Lab 02 - Google Docs

AirForShare.com - Virtual Desk

+

eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#LaunchInstances:

YouTube

aws

Search

[Alt+S]

Europe (Stockholm)

Account ID: 9965-0109-3079

hafsa_01

EC2 > Instances > Launch an instance

Application and OS Images (Amazon Machine Image)

Info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

Recents

Quick Start

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Debian

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Ubuntu Server 24.04 LTS (HVM), SSD Volume Type

Free tier eligible

ami-0a716d3f3b16d290c (64-bit (x86)) / ami-0cab1941a8a08b817 (64-bit (Arm))

Virtualization: hvm ENA enabled: true Root device type: ebs

Description

Ubuntu Server 24.04 LTS (HVM),EBS General Purpose (SSD) Volume Type. Support available from Canonical (<http://www.ubuntu.com/cloud/services>).

Summary

Number of instances

Info

1

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd6...read more

ami-0a716d3f3b16d290c

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

CloudShell

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Search

1:01 PM

10/18/2025

eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#LaunchInstances:

Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

server-1-key Create new key pair

Network settings Info Edit

Network Info

vpc-08f7b340e0705d05b

Subnet Info

No preference (Default subnet in any availability zone)

Auto-assign public IP Info

Enable

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group Select existing security group

Common security groups Info

Select security groups

Summary

Number of instances Info

1

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd6...read more

ami-0a716d3f3b16d290c

Virtual server type (instance type)

t3.micro

Firewall (security group)

Security-group-for-ec2

Storage (volumes)

1 volume(s) - 8 GiB

Cancel Launch instance Preview code

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Search

1:02 PM 10/18/2025

eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#LaunchInstances:

Network settings Info Edit

Network Info

vpc-08f7b340e0705d05b

Subnet Info

No preference (Default subnet in any availability zone)

Auto-assign public IP Info

Enable

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group Select existing security group

Common security groups Info

Select security groups

Security-group-for-ec2 sg-0f9ffdfa55cc7a8d0 X

VPC: vpc-08f7b340e0705d05b

Compare security group rules

Summary

Number of instances Info

1

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd6...read more

ami-0a716d3f3b16d290c

Virtual server type (instance type)

t3.micro

Firewall (security group)

Security-group-for-ec2

Storage (volumes)

1 volume(s) - 8 GiB

Cancel Launch instance Preview code

Configure storage Info Advanced

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Search

1:02 PM 10/18/2025

Launch an instance | EC2 | eu-north-1

eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#LaunchInstances:

EC2 > Instances > Launch an instance

Select

User data - optional [Info](#)

Upload a file with your user data or enter it in the field.

[Choose file](#)

```
#!/bin/bash
yes | sudo apt update
yes | sudo apt install apache2 -y

HOSTNAME=$(hostname)
IP_ADDRESS=$(hostname -I | awk '{print $1}')
echo "<h1>IP Address of this server is</h1><p><strong>$IP_ADDRESS</strong></p>" | sudo tee /var/www/html/index.html
sudo systemctl restart apache2
```

☐ User data has already been base64 encoded

Summary

Number of instances [Info](#)

1

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd64...[read more](#)

ami-0a716d3f3b16d290c

Virtual server type (instance type)

t3.micro

Firewall (security group)

Security-group-for-ec2

Storage (volumes)

1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

CloudShell Feedback

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1:03 PM 10/18/2025

Launch an instance | EC2 | eu-north-1

eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#LaunchInstances:

EC2 > Instances > Launch an instance

Success

Successfully initiated launch of instance (i-034ab82d35d7fb5b0)

► Launch log

Next Steps

What would you like to do next with this instance, for example "create alarm" or "create backup"

Create billing usage alerts

To manage costs and avoid surprise bills, set up email notifications for billing usage thresholds.

[Create billing alerts](#)

Connect to your instance

Once your instance is running, log into it from your local computer.

[Connect to instance](#) [Learn more](#)

Connect an RDS database

Configure the connection between an EC2 instance and a database to allow traffic flow between them.

[Connect an RDS database](#) [Create a new RDS database](#) [Learn more](#)

Create EBS snapshot policy

Create a policy that automates the creation, retention, and deletion of EBS snapshots

[Create EBS snapshot policy](#)

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1:04 PM 10/18/2025

Instances | EC2 | eu-north-1

eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#Instances:

EC2 > Instances

Instances (2) Info

Find Instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS
Server-1	i-0fd8ba8cac3f59b40	Running	t3.micro	3/3 checks passed	View alarms +	eu-north-1b	ec2-13-60-228-245.e
server-2	i-034ab82d35d7fb5b0	Running	t3.micro	Initializing	View alarms +	eu-north-1b	ec2-51-21-168-242.e

Select an instance

//Open server-01 and copy public IPv4

Instance details | EC2 | eu-north-1

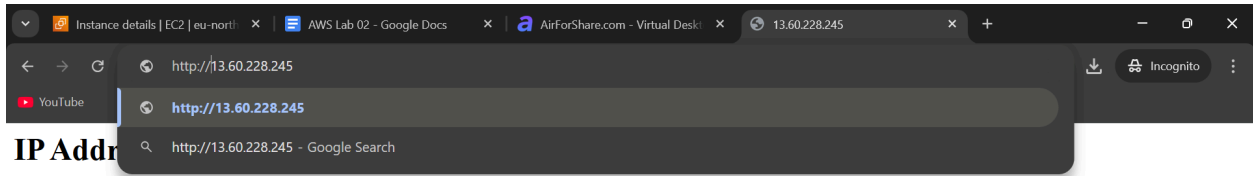
eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#InstanceDetails:instanceId=i-0fd8ba8cac3f59b40

EC2 > Instances > i-0fd8ba8cac3f59b40

Instance summary for i-0fd8ba8cac3f59b40 (Server-1) Info

Updated less than a minute ago

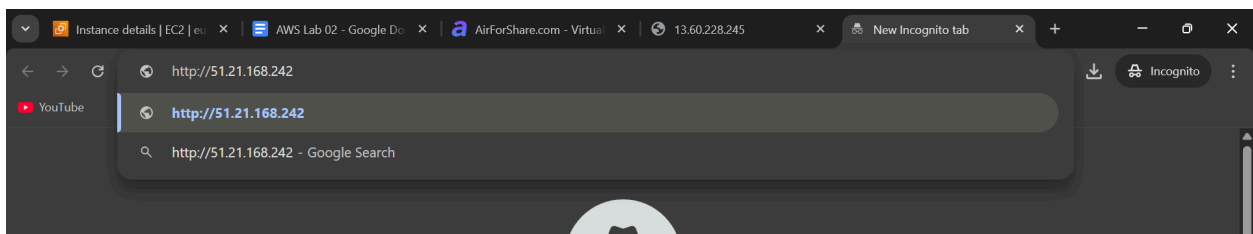
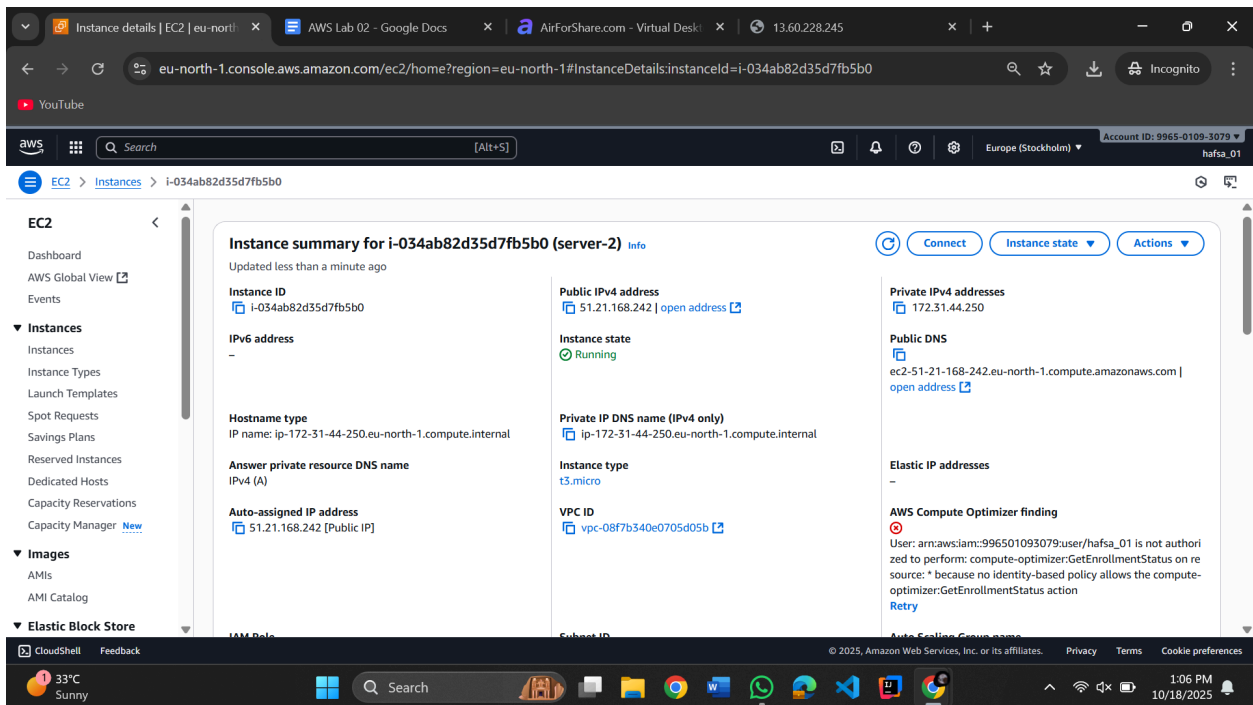
Instance ID i-0fd8ba8cac3f59b40	Public IPv4 address 13.60.228.245 open address	Private IPv4 addresses 172.31.35.135
IPv6 address -	Instance state Running	Public DNS ec2-13-60-228-245.eu-north-1.compute.amazonaws.com open address
Hostname type IP name: ip-172-31-35-135.eu-north-1.compute.internal	Private IP DNS name (IPv4 only) ip-172-31-35-135.eu-north-1.compute.internal	Elastic IP addresses -
Answer private resource DNS name IPv4 (A)	Instance type t3.micro	AWS Compute Optimizer finding User: arn:aws:iam::996501093079:user/hafsa_01 is not authorized to perform: compute-optimizer:GetEnrollmentStatus on resource: * because no identity-based policy allows the compute-optimizer:GetEnrollmentStatus action Retry
Auto-assigned IP address 13.60.228.245 [Public IP]	VPC ID vpc-08f7b340e0705d05b	



IP Address

172.31.35.135

//Do the same with server-2 -> Copy public IPv4



IP Address of this server is

172.31.44.250

//Making target group now



 Search



[EC2](#) > [Instances](#) > i-034

Volumes

Snapshots

Lifecycle Manager

 **Network & Security**

Security Groups

Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

 **Load Balancing**

Load Balancers

Target Groups

Trust Stores

 **Auto Scaling**

Auto Scaling Groups

Settings

//Create target group

Target groups | EC2 | eu-north-1

eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#TargetGroups:

Account ID: 9965-0109-3079 | hafsa_01

EC2 > Target groups

Volumes
Snapshots
Lifecycle Manager

▼ Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces

▼ Load Balancing
Load Balancers
Target Groups
Trust Stores

▼ Auto Scaling
Auto Scaling Groups

Target groups info

Filter target groups

Name	ARN	Port	Protocol	Target type	Load balancer	VPC ID
No target groups						
You don't have any target groups in eu-north-1						

Create target group

0 target groups selected

Select a target group above.

Step 1 Create target group

eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#CreateTargetGroup:

Account ID: 9965-0109-3079 | hafsa_01

EC2 > Target groups > Create target group

Register targets

Basic configuration

Settings in this section can't be changed after the target group is created.

Choose a target type

☒ **Instances**

- Supports load balancing to instances within a specific VPC.
- Facilitates the use of [Amazon EC2 Auto Scaling](#) to manage and scale your EC2 capacity.

☐ IP addresses

- Supports load balancing to VPC and on-premises resources.
- Facilitates routing to multiple IP addresses and network interfaces on the same instance.
- Offers flexibility with microservice based architectures, simplifying inter-application communication.
- Supports IPv6 targets, enabling end-to-end IPv6 communication, and IPv4-to-IPv6 NAT.

☐ Lambda function

- Facilitates routing to a single Lambda function.
- Accessible to Application Load Balancers only.

☐ Application Load Balancer

- Offers the flexibility for a Network Load Balancer to accept and route TCP requests within a specific VPC.
- Facilitates using static IP addresses and PrivateLink with an Application Load Balancer.

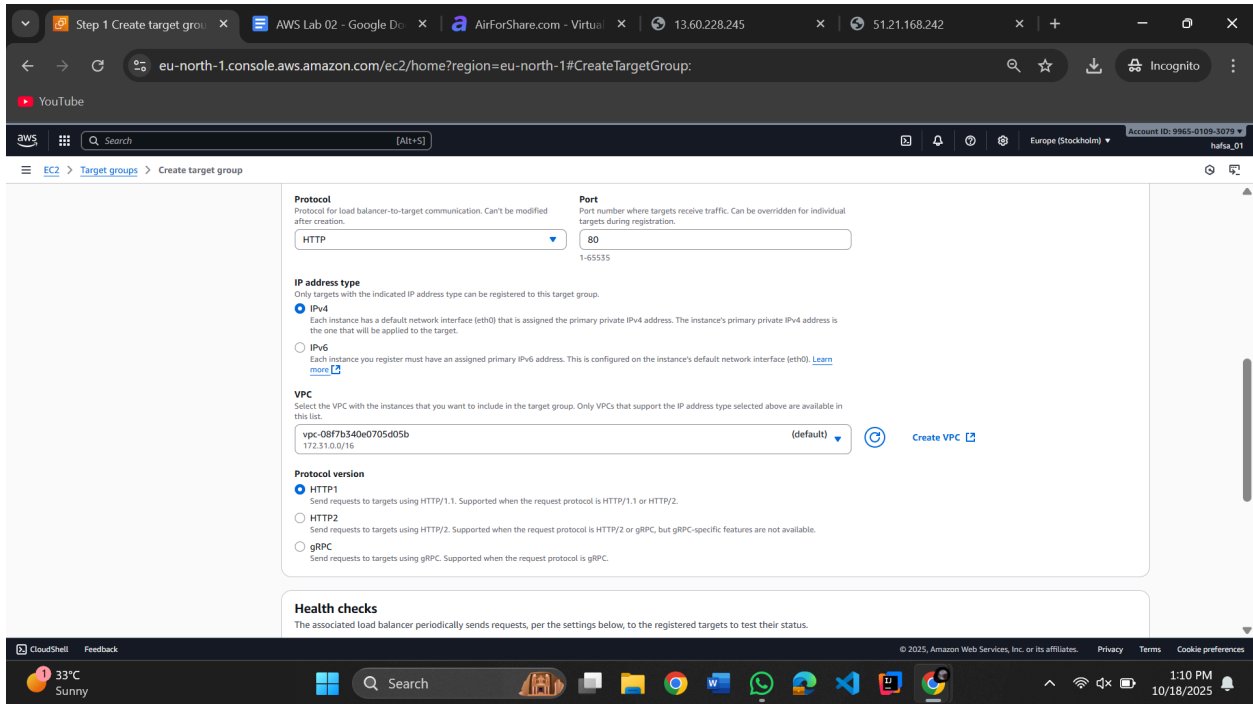
Target group name

Target-group-for-LB

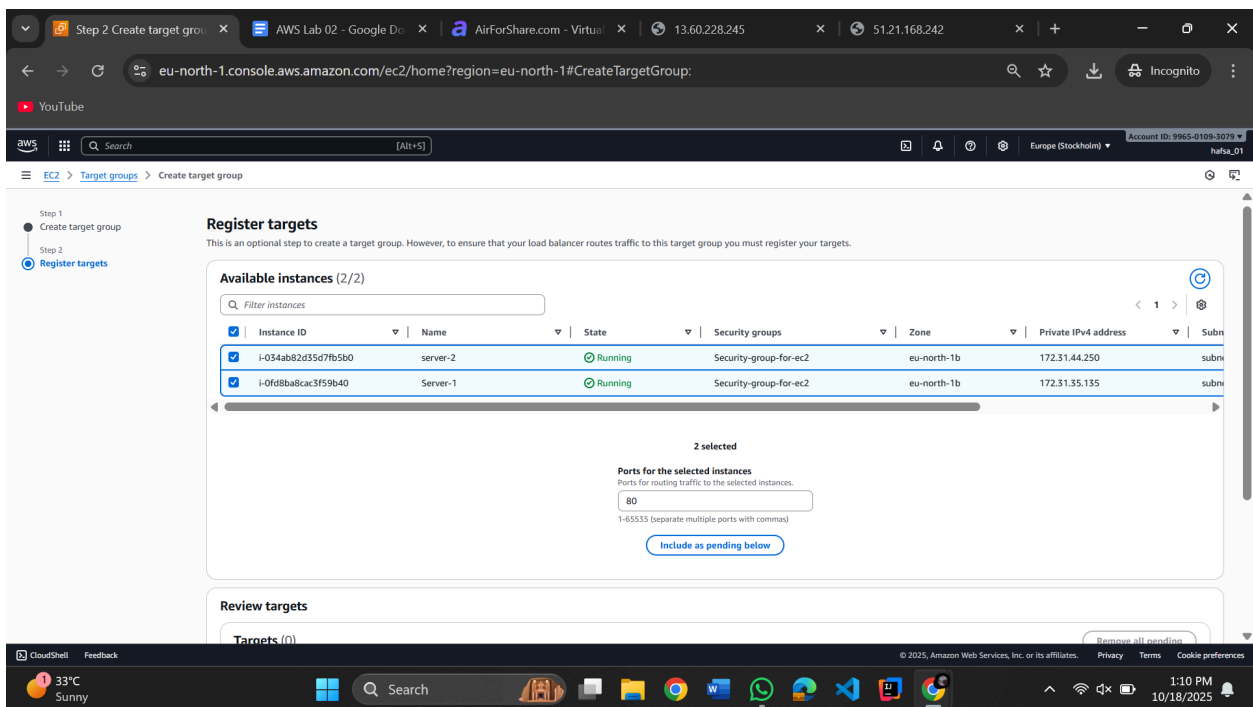
A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

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33°C Sunny 1:10 PM 10/18/2025



//Click next



Click include as pending below (Port should be 80)

Review targets

Targets (2)

Show only pending

Remove all pending

Instance ID	Name	Port	State	Security groups	Zone	Private IPv4 address	Subnet ID	Launch time
i-034ab82d35d7fb5b0	server-2	80	Running	Security-group-for-ec2	eu-north-1b	172.31.44.250	subnet-088f2244939c7056a	October 18, 2025, 13:04 (UTC+05:00)
i-0fd8ba8cac3f59b40	Server-1	80	Running	Security-group-for-ec2	eu-north-1b	172.31.35.135	subnet-088f2244939c7056a	October 18, 2025, 12:59 (UTC+05:00)

2 pending

Cancel

Previous

Create target group

Target group details | EC2

AWS Lab 02 - Google D...

AirForShare.com - Virtual

13.60.228.245

51.21.168.242

eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#TargetGroup:targetGroupArn=arn:aws:elasticloadbalancing:eu-north...

Incognito

EC2 > Target groups > Target-group-for-LB

Capacity manager

Images

Elastic Block Store

Network & Security

Load Balancing

Auto Scaling

Successfully created the target group: **Target-group-for-LB**. Anomaly detection is automatically applied to all registered targets. Results can be viewed in the **Targets** tab.

Target-group-for-LB

Actions

Details

Target type

Instance

Protocol : Port

HTTP: 80

IP address type

IPv4

Load balancer

None associated

Protocol version

HTTP1

VPC

vpc-08f7b340e0705d05b

2 Total targets

0 Healthy

0 Unhealthy

2 Unused

0 Initial

0 Draining

0 Anomalous

Distribution of targets by Availability Zone (AZ)

Select values in this table to see corresponding filters applied to the Registered targets table below.

Targets

Monitoring

Health checks

Attributes

Tags

Registered targets (2)

Anomaly mitigation: Not applicable

Deregister

Register targets

Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.

CloudShell

Feedback

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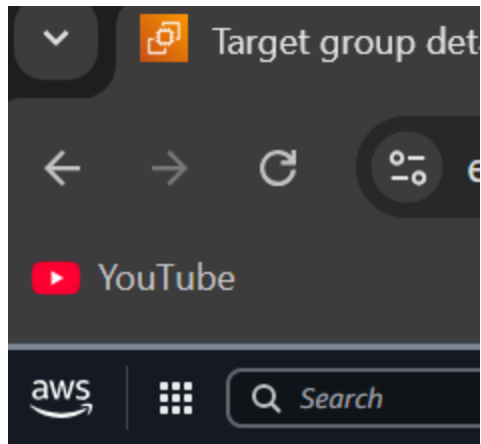
33°C Sunny

Search

11:11 PM

10/18/2025

//Open Load balancer



 [EC2](#) > [Target groups](#) > T
Capacity Manager [New](#)

▼ **Images**

AMIs
AMI Catalog

▼ **Elastic Block Store**

Volumes
Snapshots
Lifecycle Manager

▼ **Network & Security**

Security Groups
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▼ **Load Balancing**

[Load Balancers](#)
Target Groups
Trust Stores

▼ **Auto Scaling**

Auto Scaling Groups

Settings

Create load balancer

The screenshot shows the AWS Management Console for the 'eu-north-1' region. The left-hand navigation pane is open, showing the 'Load Balancing' section with 'Load Balancers' selected. The main content area displays the 'Load balancers' page, which includes a search bar, a table of existing load balancers (currently empty), and a 'Create load balancer' button. A notification banner at the top introduces 'URL rewrite for Application Load Balancer'.

//Create application balancer -> Click

The screenshot shows the 'Compare and select load balancer type' wizard in the AWS Management Console. The wizard presents three options: 'Application Load Balancer', 'Network Load Balancer', and 'Gateway Load Balancer'. Each option includes a diagram illustrating its architecture and a brief description of when to choose it. The 'Application Load Balancer' option is highlighted, showing a diagram with a laptop, an ALB, and target instances. The 'Network Load Balancer' option shows a diagram with a laptop, a VPC, an NLB, and target instances. The 'Gateway Load Balancer' option shows a diagram with a laptop, a GWLB, and target instances.

Create application load balancer | AWS Lab 02 - Google Drive | AirForShare.com - Virtual | 13.60.228.245 | 51.21.168.242 | eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#CreateALBWizard:

YouTube

AWS

EC2 > Load balancers > Create Application Load Balancer

Create Application Load Balancer [info](#)

The Application Load Balancer distributes incoming HTTP and HTTPS traffic across multiple targets such as Amazon EC2 instances, microservices, and containers, based on request attributes. When the load balancer receives a connection request, it evaluates the listener rules in priority order to determine which rule to apply, and if applicable, it selects a target from the target group for the rule action.

► How Application Load Balancers work

Basic configuration

Load balancer name
Name must be unique within your AWS account and can't be changed after the load balancer is created.

My-load-balancer

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme | [Info](#)
Scheme can't be changed after the load balancer is created.

☒ **Internet-facing**

- Serves internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

☐ **Internal**

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.
- Compatible with the IPv4 and Dualstack IP address types.

Load balancer IP address type | [Info](#)
Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.

☒ **IPv4**
Includes only IPv4 addresses.

☐ **Dualstack**
Includes IPv4 and IPv6 addresses.

☐ **Dualstack without public IPv4**
Includes a public IPv6 address, and private IPv4 and IPv6 addresses. Compatible with internet-facing load balancers only.

CloudShell Feedback

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NFO - CFC In 4 hours

Search

1:16 PM 10/18/2025

Create application load balancer | AWS Lab 02 - Google Drive | AirForShare.com - Virtual | 13.60.228.245 | 51.21.168.242 | eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#CreateALBWizard:

YouTube

AWS

EC2 > Load balancers > Create Application Load Balancer

Network mapping [info](#)

The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC | [Info](#)
The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view [target groups](#).

vpc-08f7b340e0705d05b (default) [Create VPC](#)

IP pools | [Info](#)
You can optionally choose to configure an IPAM pool as the preferred source for your load balancers IP addresses. Create or view Pools in the [Amazon VPC IP Address Manager console](#).

☐ **Use IPAM pool for public IPv4 addresses**
The IPAM pool you choose will be the preferred source of public IPv4 addresses. If the pool is depleted IPv4 addresses will be assigned by AWS.

Availability Zones and subnets | [Info](#)
Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

☒ **eu-north-1a (eu1-a21)**

Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-067d1ae682ca5e233
IPv4 subnet CIDR: 172.31.0.0/20

☒ **eu-north-1b (eu1-a22)**

Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-088f2244939c7056a
IPv4 subnet CIDR: 172.31.32.0/20

☒ **eu-north-1c (eu1-a23)**

Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-01e4804b7d75b2a61
IPv4 subnet CIDR: 172.31.0.0/20

CloudShell Feedback

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NFO - CFC In 4 hours

Search

1:17 PM 10/18/2025

eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#CreateALBWizard:

Security groups info

A security group is a set of firewall rules that control the traffic to your load balancer. Select an existing security group, or you can [create a new security group](#).

Security groups

Select up to 5 security groups

Security-group-for-load-balancer sg-09a564473011c7dfa VPC: vpc-08f7b340e0705d05b

default sg-093fe7f396203778 VPC: vpc-08f7b340e0705d05b

Listeners and routing info

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

Listener HTTP:80

Protocol HTTP Port 80

Default action info

The default action is used if no other rules apply. Choose the default action for traffic on this listener.

Routing action

Forward to target groups Redirect to URL Return fixed response

Forward to target group info

Choose a target group and specify routing weight or [create target group](#).

Target group

Target-group-for-LB Target type: Instance, IPv4 Target stickiness: Off HTTP

Weight 1 Percent 100%

Target group stickiness info

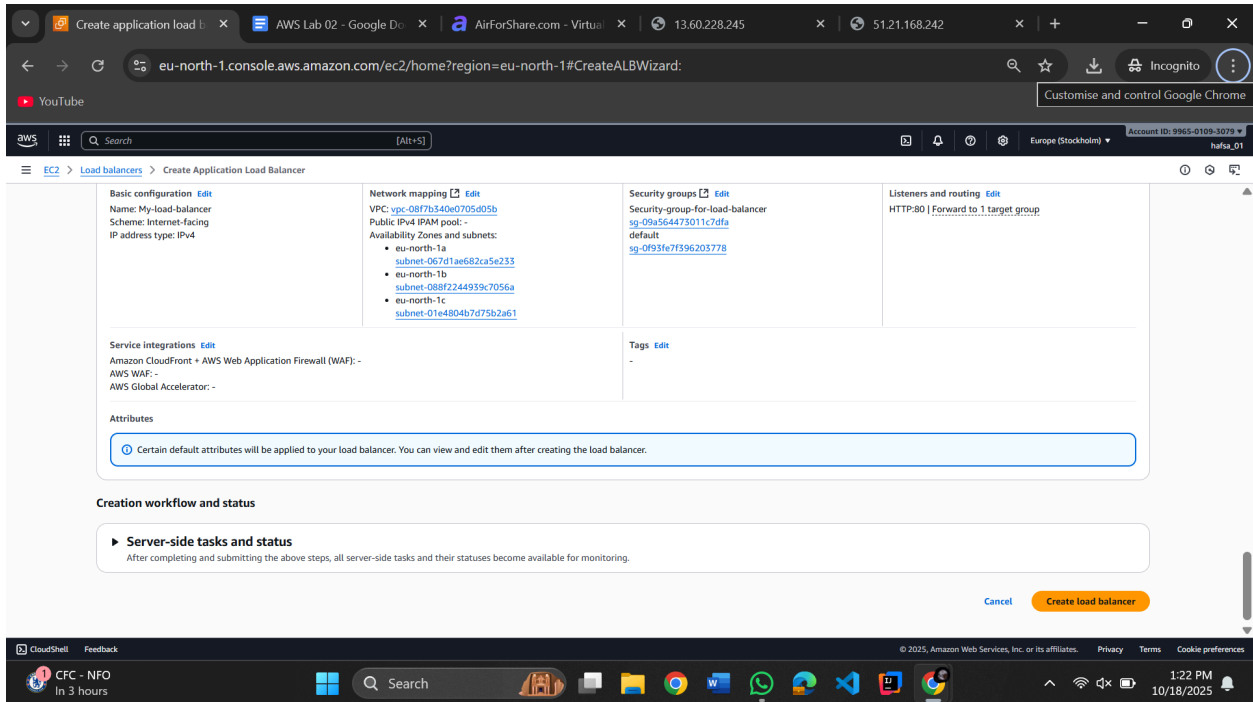
Enables the load balancer to bind a user's session to a specific target group. To use stickiness the client must support cookies. If you want to bind a user's session to a specific target, turn on the Target Group attribute Stickiness.

Turn on target group stickiness

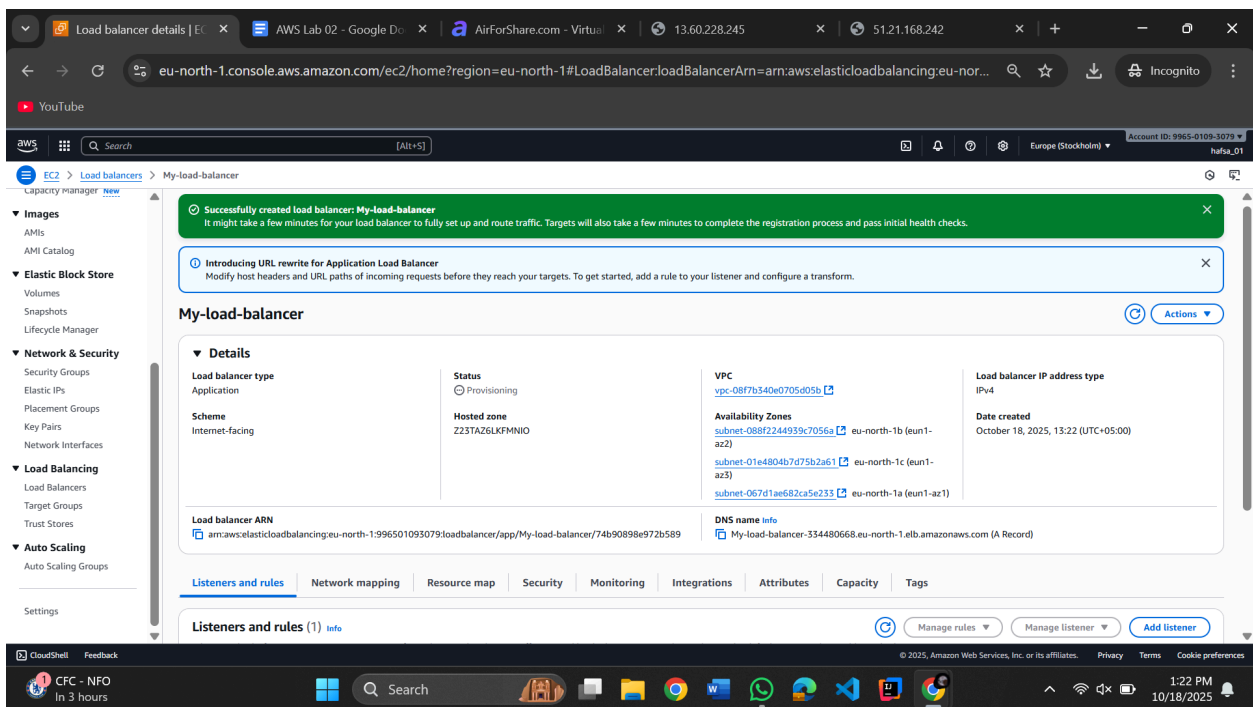
Listener tags - optional

Consider adding tags to your listener. Tags enable you to categorize your AWS resources so you can more easily manage them.

(Since we have one target group, we will choose the weight “1”)

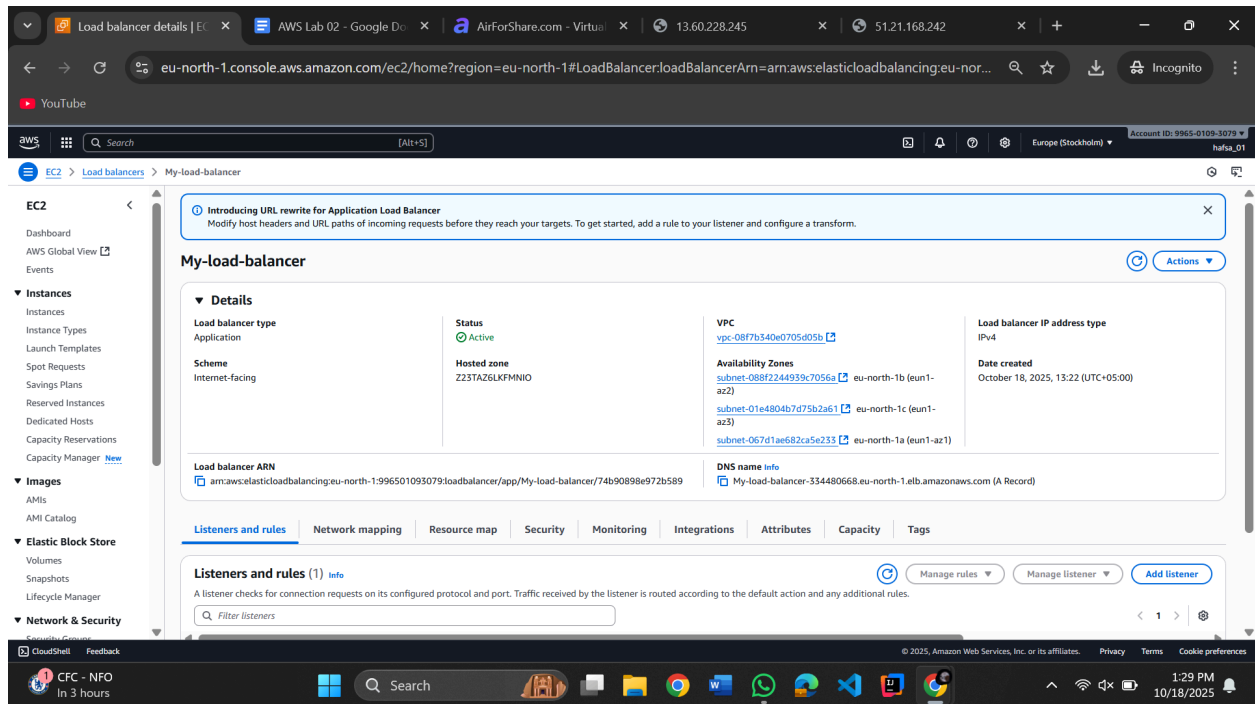


Create it!!!

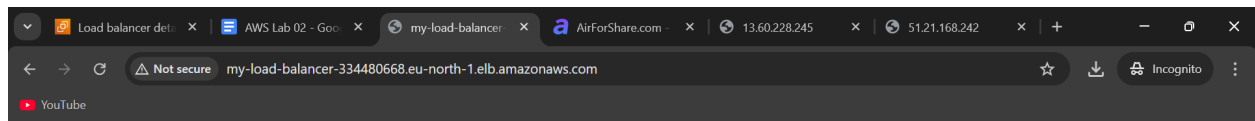
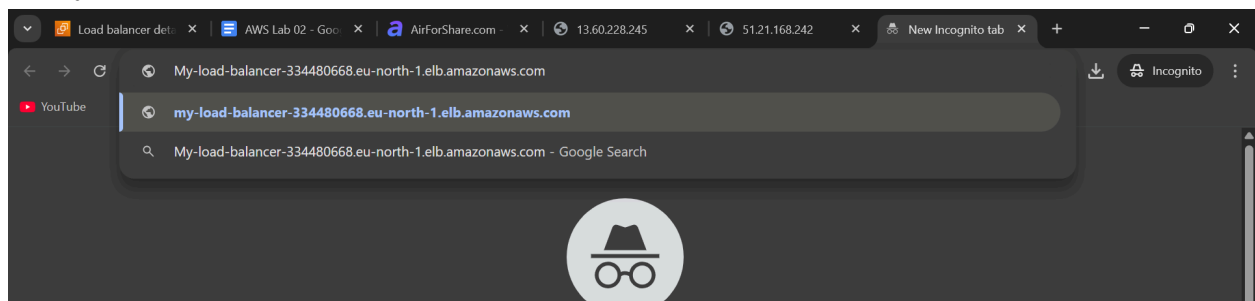


Whehehe

//Refresh (Status will change from provisioning to active)

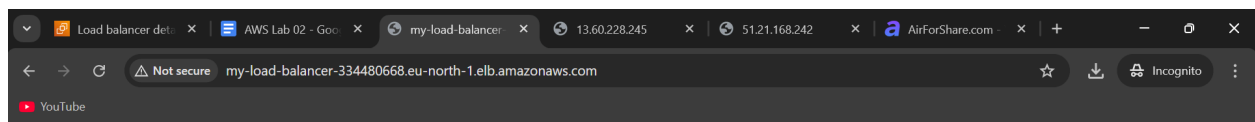


//Copy DNS Name



IP Address of this server is

172.31.44.250

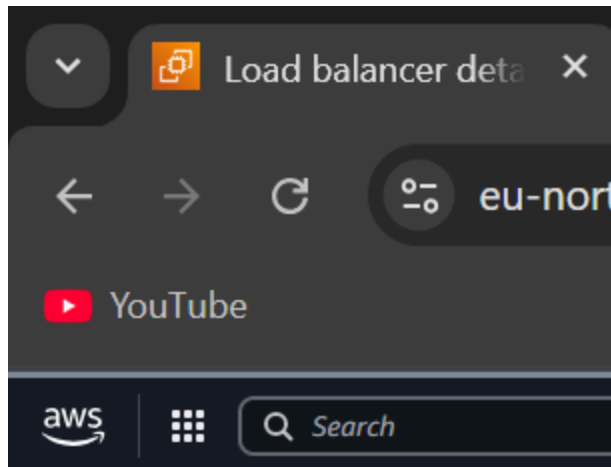


IP Address of this server is

172.31.35.135

(Refresh the page. The load balancer DNS will connect to either one of the EC2. I will only work once the status is ACTIVE!)

```
//Terminating the load balancer
```



EC2 > [Load balancers](#) > My-load-l

Capacity Manager [New](#)

- ▼ **Images**
 - AMIs
 - AMI Catalog
- ▼ **Elastic Block Store**
 - Volumes
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 - Network Interfaces
- ▼ **Load Balancing**
 - [Load Balancers](#)
 - Target Groups
 - Trust Stores
- ▼ **Auto Scaling**
 - Auto Scaling Groups

Settings

Introducing URL rewrite for Application Load Balancer
Modify host headers and URL paths of incoming requests before they reach your targets. To get started, add a rule to your listener and configure a transform.

Load balancers (1/1)
Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

Name	State	Type	Scheme	IP address type	VPC ID	Availability Zones
My-load-balancer	Active	application	Internet-facing	IPv4	vpc-08f7b340e0705d05b	5 Availability Zones

Load balancer: My-load-balancer

Details

Load balancer type Application	Status Active	VPC vpc-08f7b340e0705d05b	Load balancer IP address type IPv4
Scheme Internet-facing	Hosted zone Z23TAZ6LKFHMNIO	Availability Zones subnet-088f2244939c7056a eu-north-1b (eu-n1-az2) subnet-01e4804b7d75b2a61 eu-north-1c (eu-n1-az3)	Date created October 18, 2025, 13:22 (UTC+05:00)

Delete load balancer

Delete load balancer **My-load-balancer** permanently? This action can't be undone.

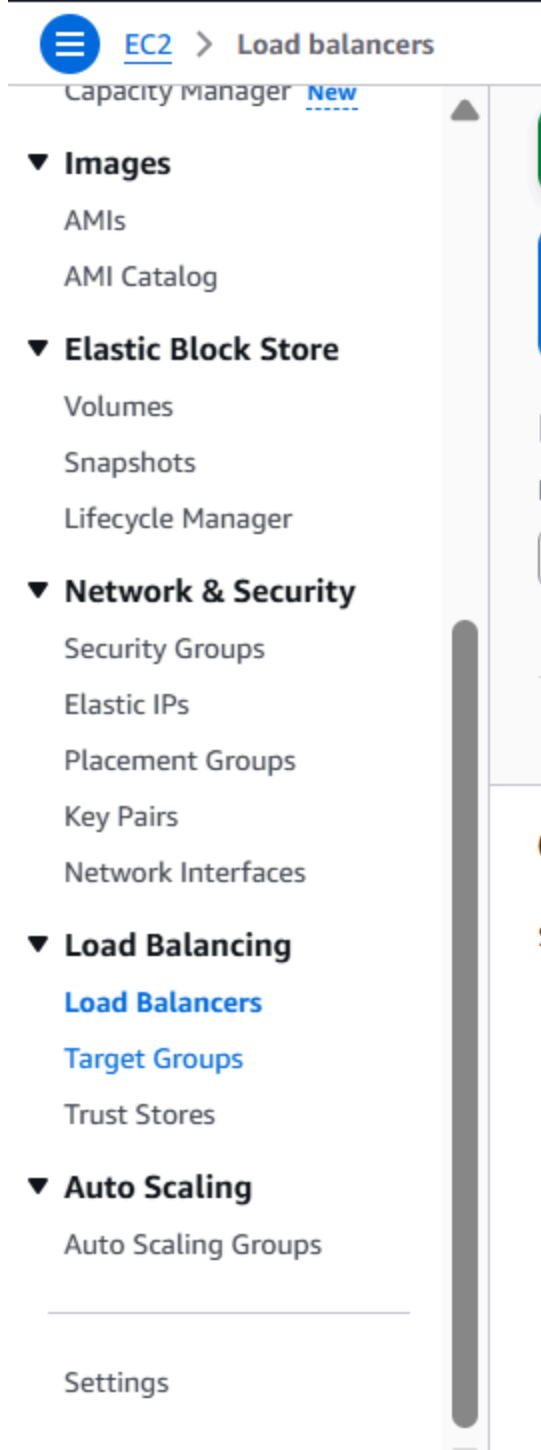
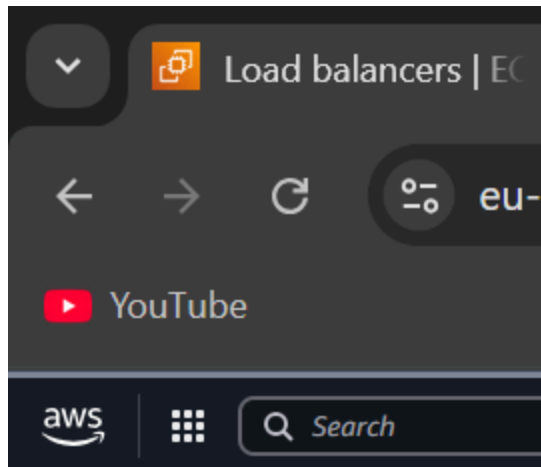
⚠ Proceeding with this action deletes the load balancer and its listeners. Target groups associated to this load balancer will become available for association to another load balancer and their registered targets remain unaffected.

To avoid accidental deletion we ask you to provide additional written consent.

Type confirm to agree.
confirm

Cancel Delete

//Terminating Target groups



Target groups | EC2

eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#TargetGroups:

Target groups (1/1)

Name	ARN	Port	Protocol	Target type	Load balancer
Target-group-for-LB	arn:aws:elasticloadbalancing:eu-north-1:996501093079:targetgroup/Target-group-for-LB/bdb57073018d7a4c	80	HTTP	Instance	None associated

Target group: Target-group-for-LB

Details

Target type: Instance

Protocol: Port: HTTP: 80

IP address type: IPv4

Load balancer: None associated

Protocol version: HTTP1

VPC: vpc-08f7b340e0705d05b

Target groups | EC2

eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#TargetGroups:

Target groups (1/1)

Name	ARN	Port	Protocol	Target type	Load balancer	VPC ID
Target-group-for-LB	arn:aws:elasticloadbalancing:eu-north-1:996501093079:targetgroup/Target-group-for-LB/bdb57073018d7a4c	80	HTTP	Instance	None associated	vpc-08f7b340e0705d05b

Target group: Target-group-for-LB

Details

Target type: Instance

Protocol: Port: HTTP: 80

IP address type: IPv4

Load balancer: None associated

Protocol version: HTTP1

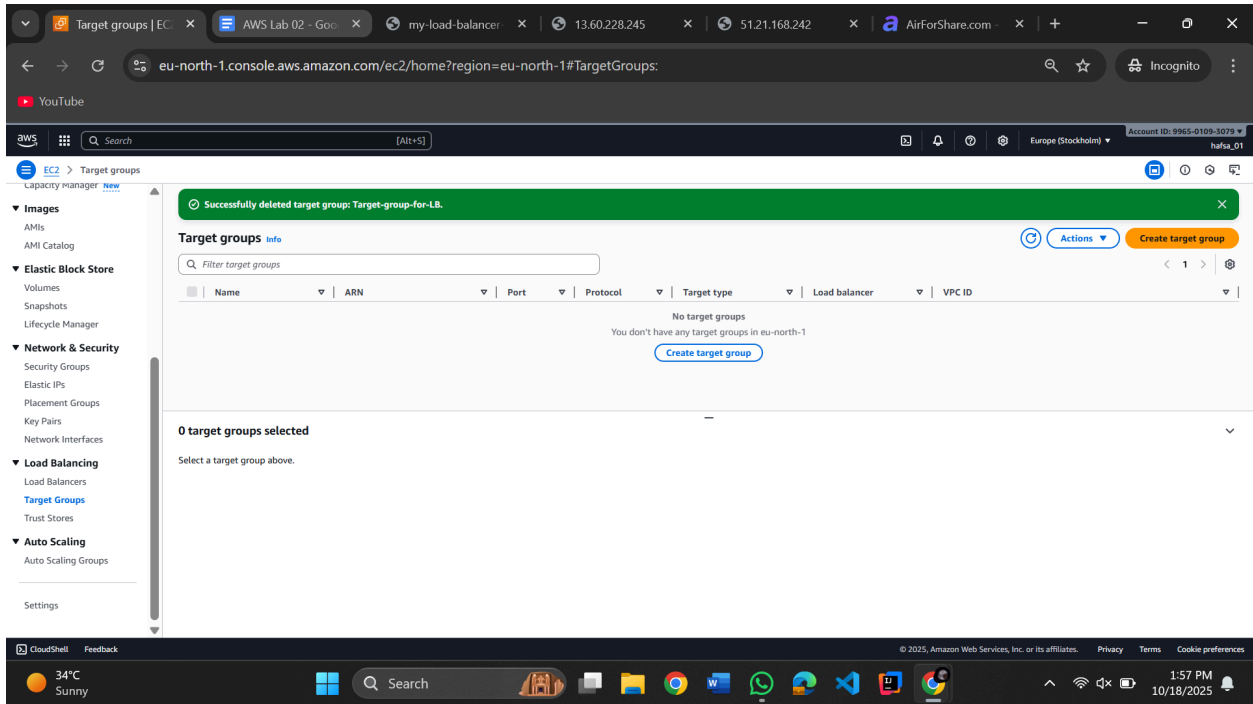
VPC: vpc-08f7b340e0705d05b

Delete target group

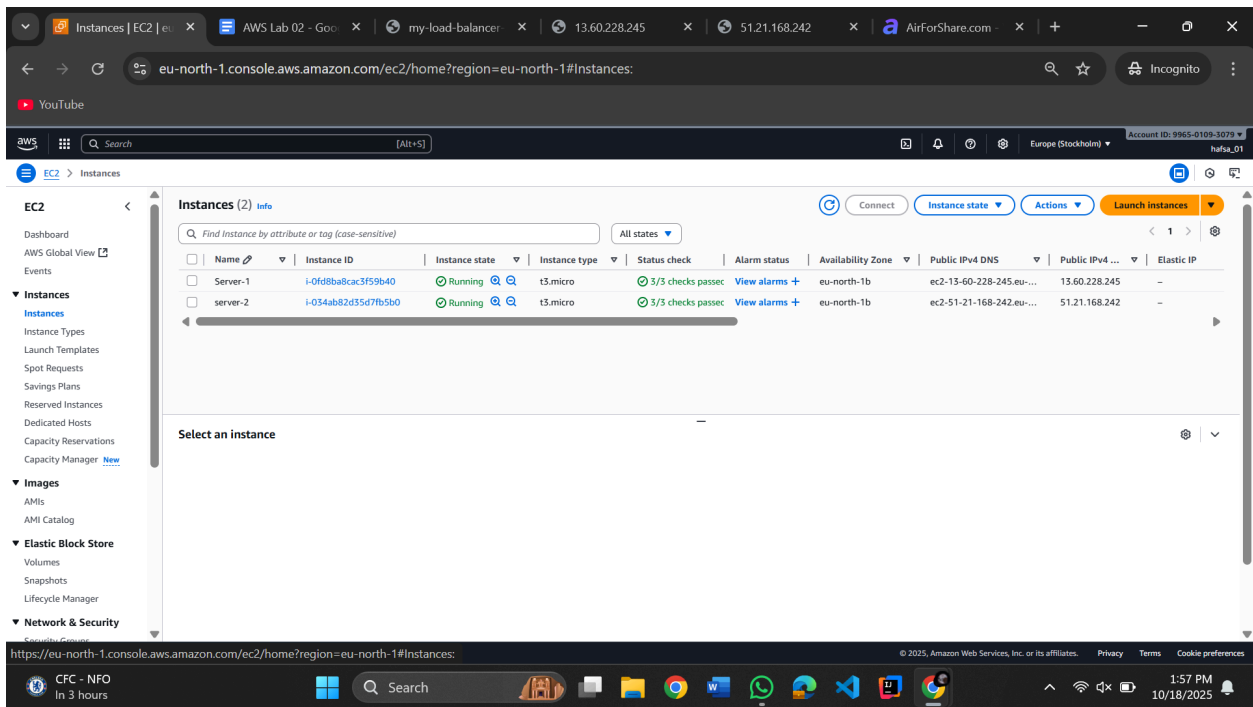
Permanently delete 1 target group. You can't undo this action.

- Target-group-for-LB

Deleting a target group deletes the group; the individual resources registered to the target group don't get deleted as a result of this action.



//Deleting



//Stopping instance

eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#Instances:

EC2 > Instances

Instances (1/2) info

Find Instance by attribute or tag (case-sensitive)

All states

Connect Instance state Actions Launch instances

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 address	Private IPv4 address	Elastic IP
Server-1	i-0f08ba8cac3f59b40	Running	t3.micro	3/3 checks passed	View alarms +	eu-north-1b	51.21.168.242	172.31.44.250	-
server-2	i-034ab82d35d7fb5b0	Running	t3.micro	3/3 checks passed	View alarms +	eu-north-1b	51.21.168.242	172.31.44.250	-

i-034ab82d35d7fb5b0 (server-2)

Details Status and alarms Monitoring Security Networking Storage Tags

Instance summary info

Instance ID i-034ab82d35d7fb5b0

Public IPv4 address 51.21.168.242 | open address

Private IPv4 addresses 172.31.44.250

Instance state Running

Public DNS ec2-51-21-168-242.eu-north-1.compute.amazonaws.com | open address

Private IP DNS name (IPv4 only) ip-172-31-44-250.eu-north-1.compute.internal

IP name: ip-172-31-44-250.eu-north-1.compute.internal

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eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#Instances:

EC2 > Instances

Successfully initiated termination (deletion) of i-034ab82d35d7fb5b0

Instances (1/2) info

Find Instance by attribute or tag (case-sensitive)

All states

Connect Instance state Actions Launch instances

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP
Server-1	i-0f08ba8cac3f59b40	Running	t3.micro	3/3 checks passed	View alarms +	eu-north-1b	ec2-13-60-228-245.eu...	13.60.228.245	-
server-2	i-034ab82d35d7fb5b0	Shutting-d...	t3.micro	3/3 checks passed	View alarms +	eu-north-1b	ec2-51-21-168-242.eu...	51.21.168.242	-

i-034ab82d35d7fb5b0 (server-2)

Details Status and alarms Monitoring Security Networking Storage Tags

Instance summary info

Instance ID i-034ab82d35d7fb5b0

Public IPv4 address 51.21.168.242 | open address

Private IPv4 addresses 172.31.44.250

Instance state Shutting-down

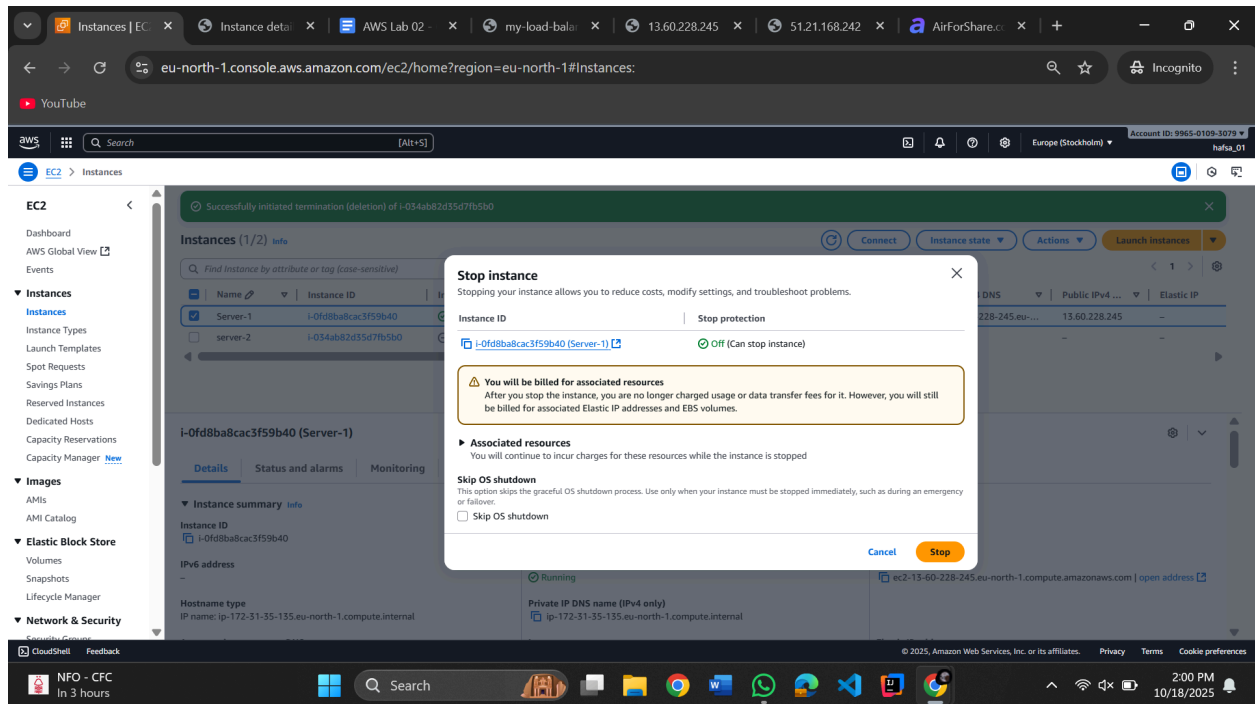
Public DNS ec2-51-21-168-242.eu-north-1.compute.amazonaws.com | open address

Private IP DNS name (IPv4 only) ip-172-31-44-250.eu-north-1.compute.internal

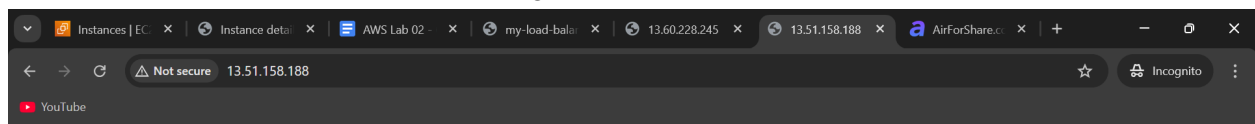
IP name: ip-172-31-44-250.eu-north-1.compute.internal

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Terminate one of them
And stop one of them (Before stopping copy their instance ID)



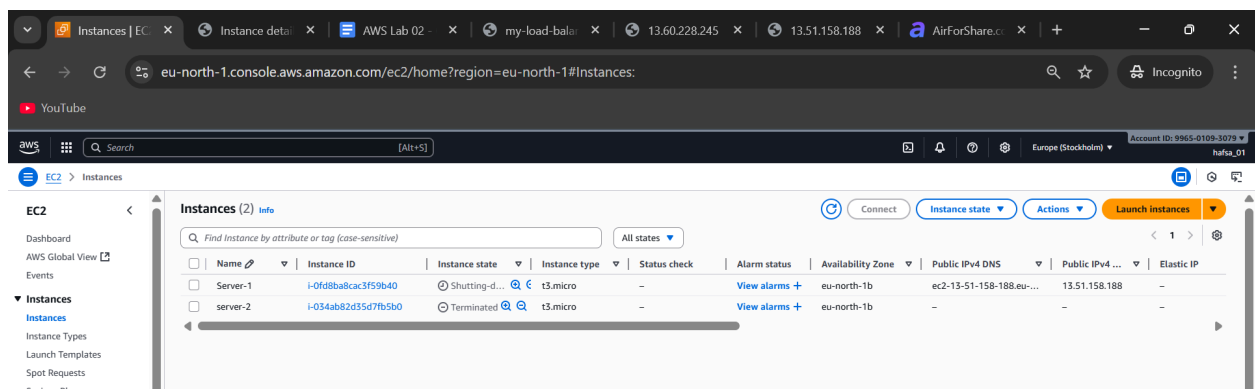
Restart the stopped instance, It will change it



IP Address of this server is

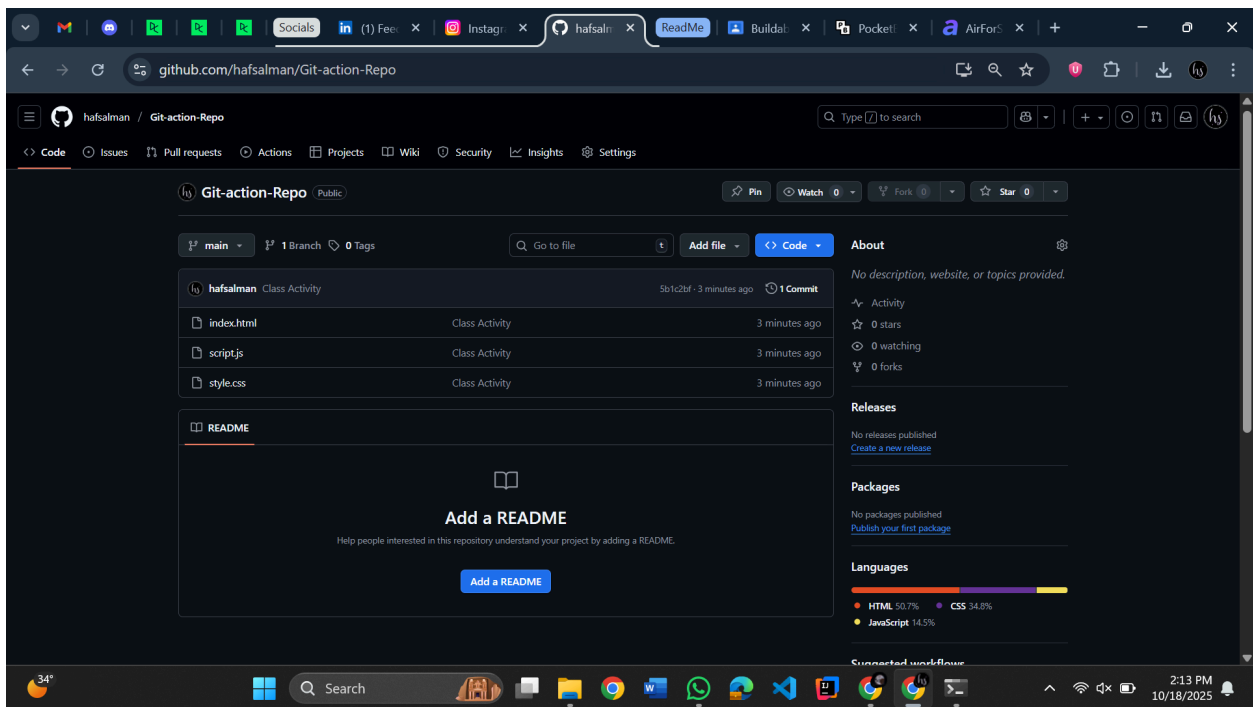
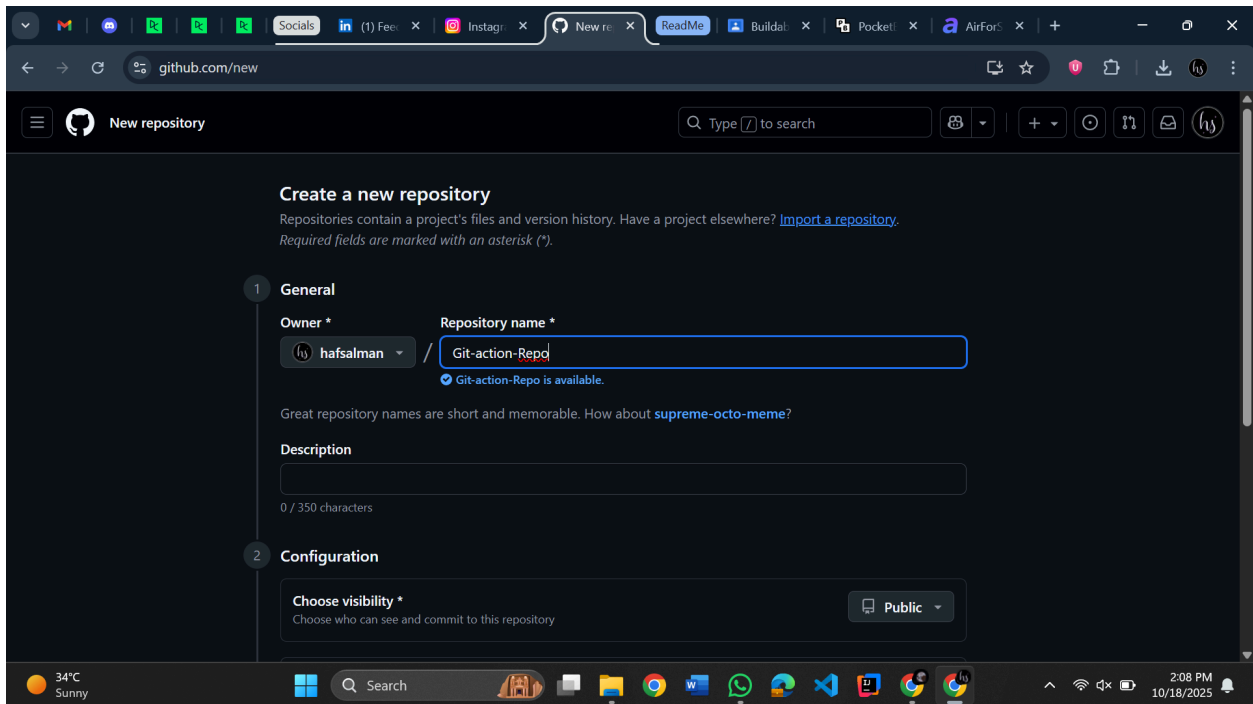
172.31.35.135

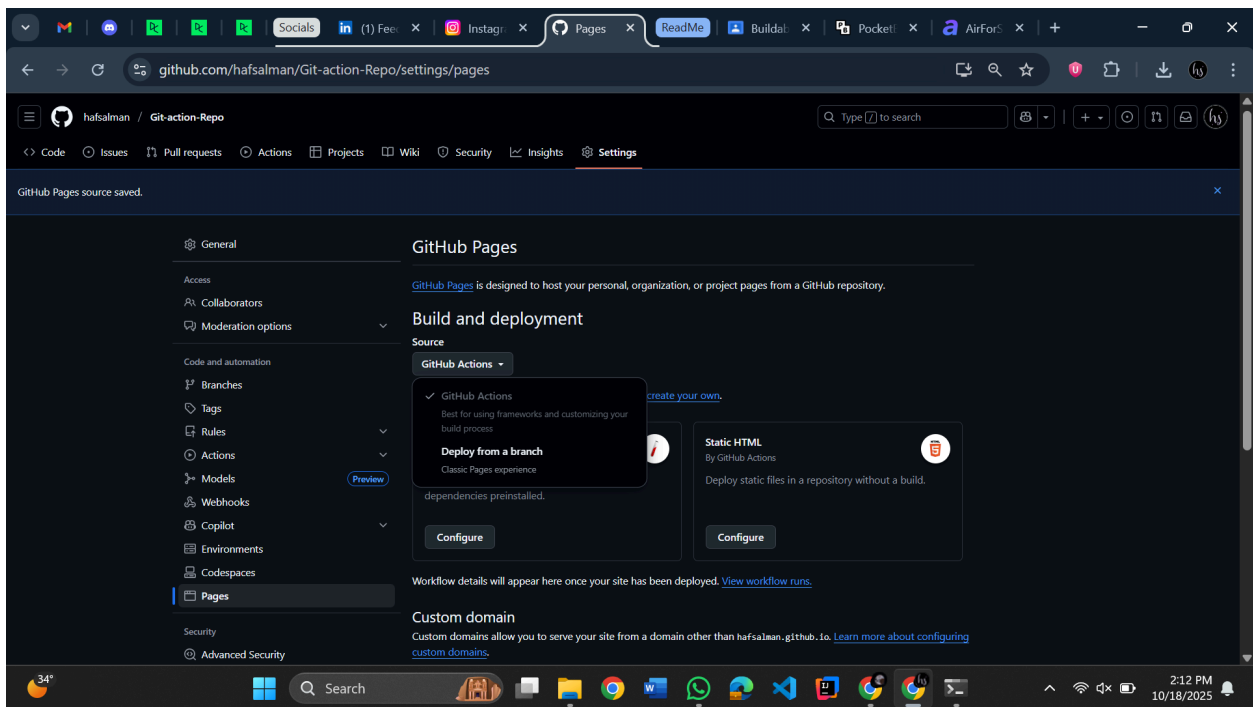
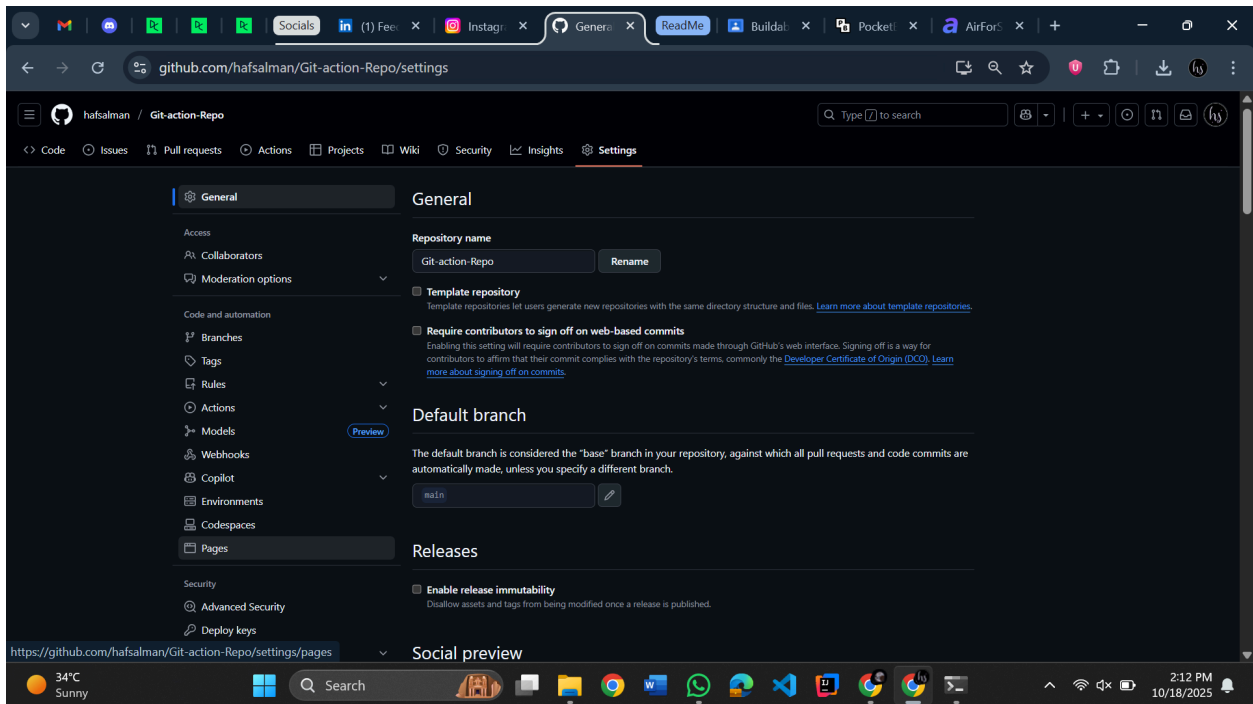
Terminate the instance



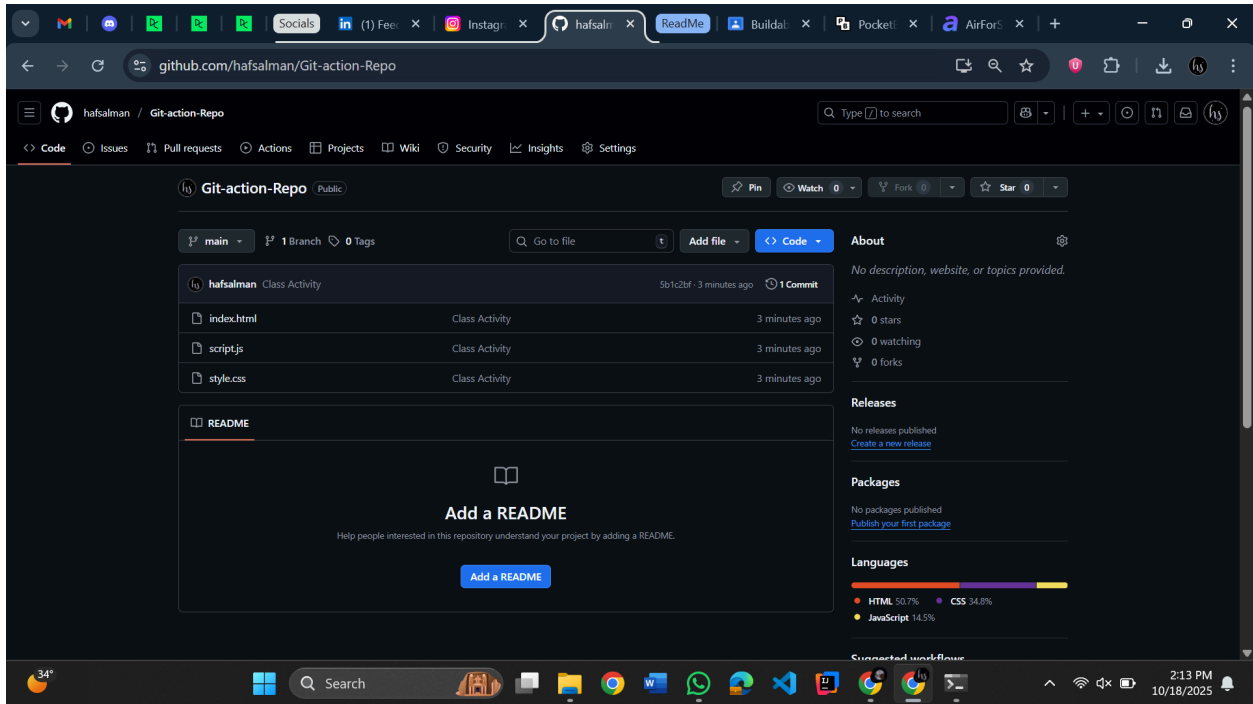
//GITHUB TIMEEEE

- Open GitHub (We will deploy application on GitHub. Next lab main we will deploy it on EC2)
- Files link (I will upload them on GitHub so you can clone them)
- Create repo

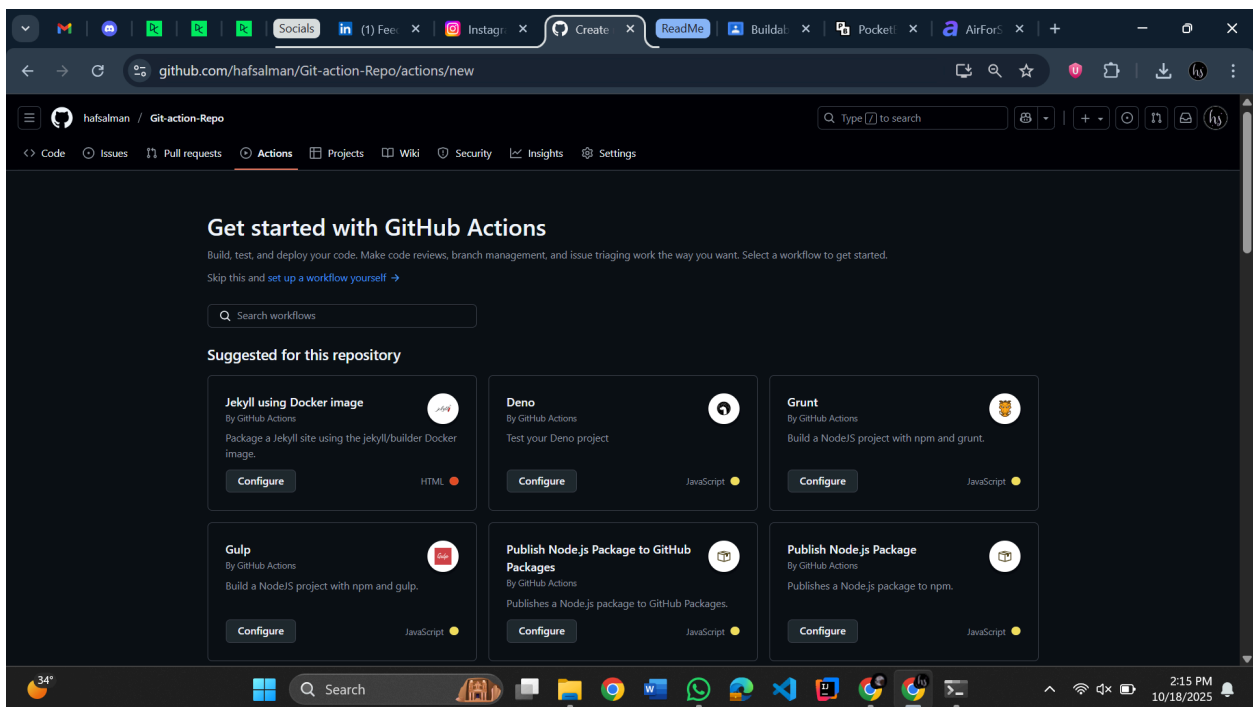




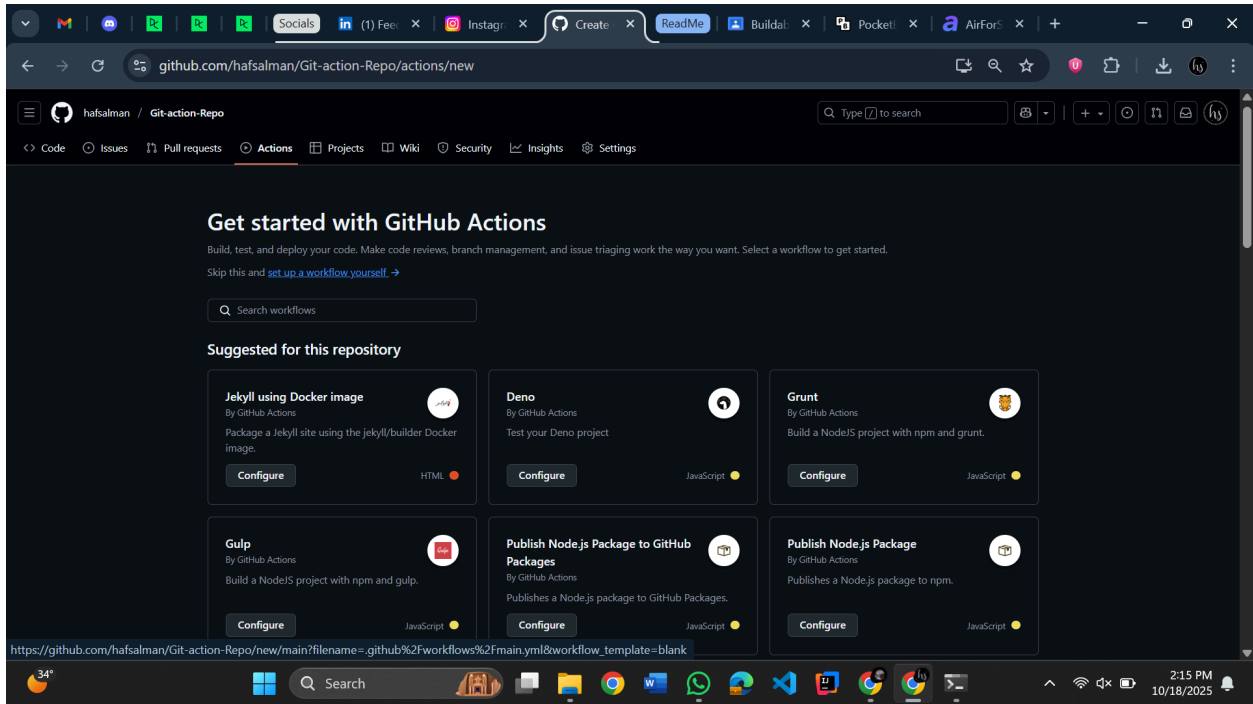
(Select GitHub Actions)
Go back to the code



Go to actions



Click "Set up a workflow yourself"



//Copy paste deploy.yml

```
name: Deploy Static Website

on:
  push:
    branches:
      - main

permissions:
  contents: read
  pages: write
  id-token: write

jobs:
  deploy:
    environment:
      name: github-pages
      url: ${ steps.deployment.outputs.page_url }
    runs-on: ubuntu-latest

    steps:
      - name: Checkout code
        uses: actions/checkout@v4
```

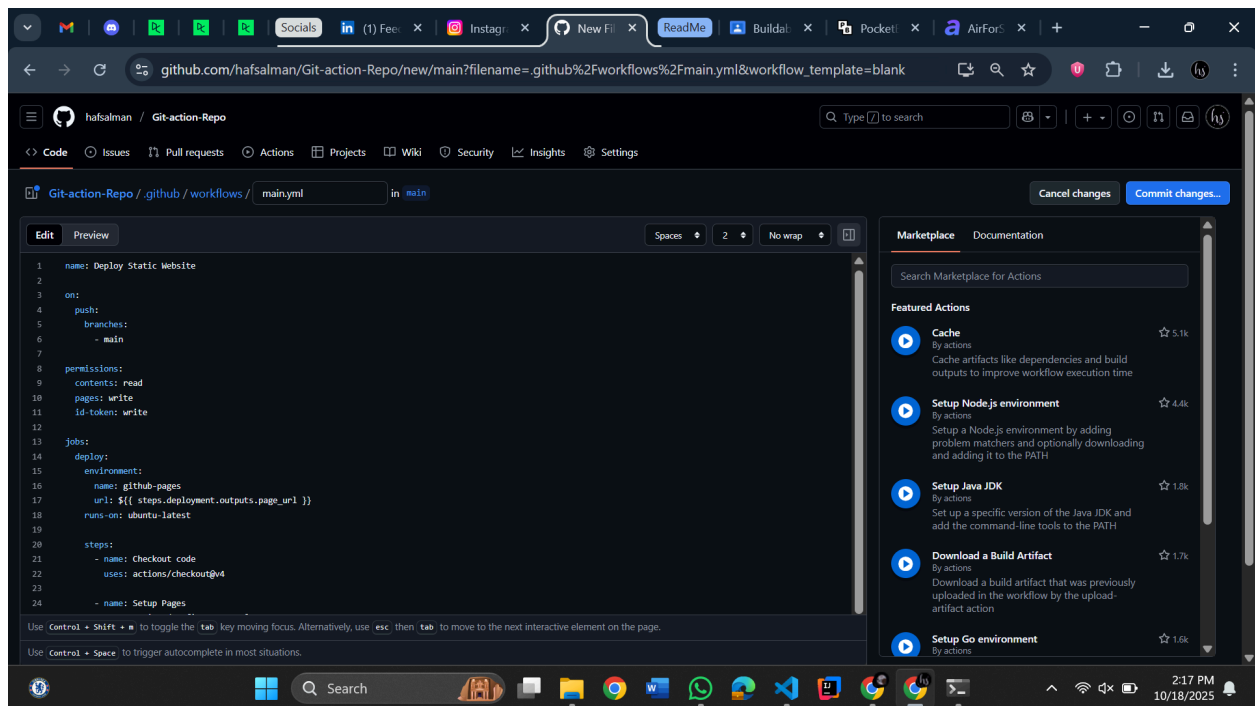
```

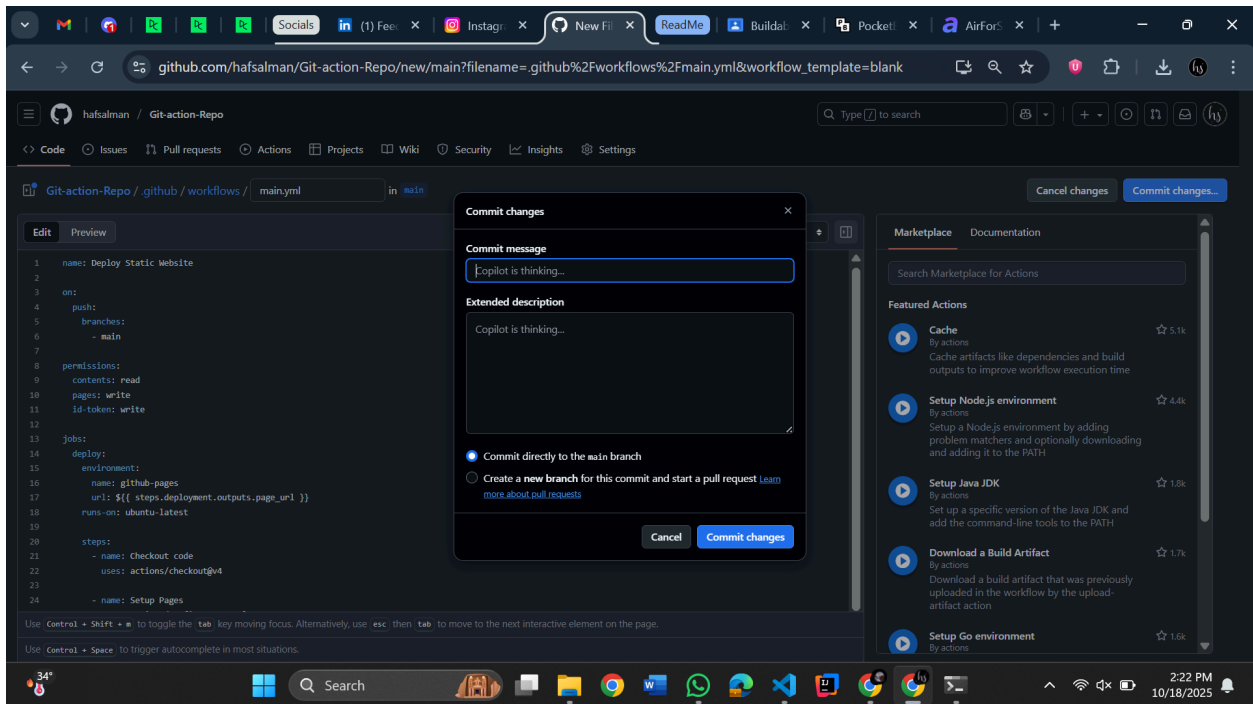
- name: Setup Pages
  uses: actions/configure-pages@v5

- name: Upload website files
  uses: actions/upload-pages-artifact@v3
  with:
    path: ./ # Path to your static files

- name: Deploy to GitHub Pages
  id: deployment
  uses: actions/deploy-pages@v4

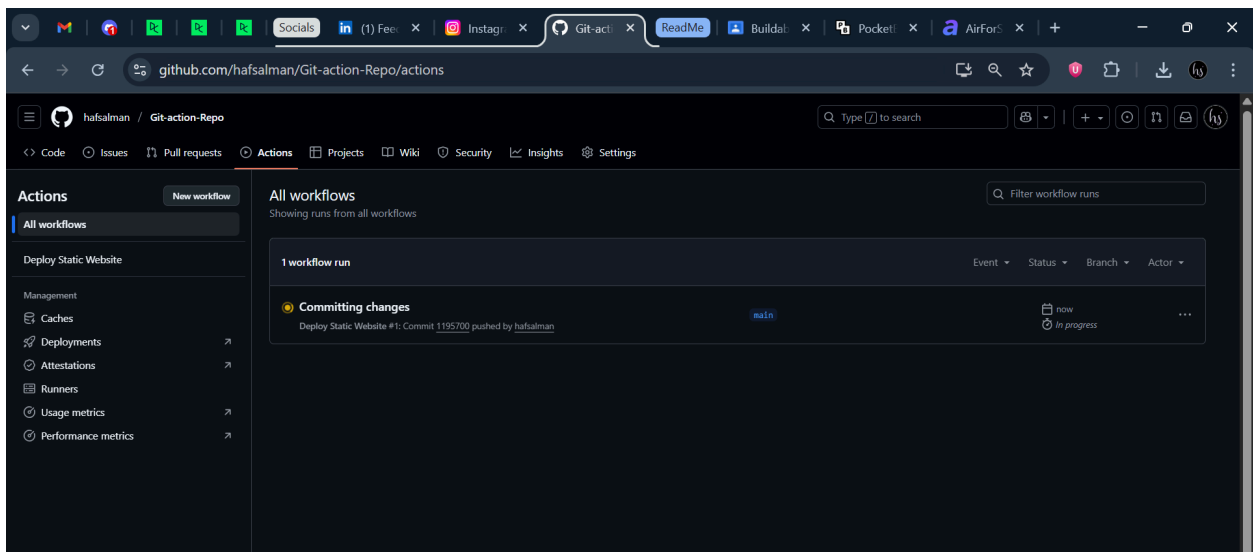
```





Commit changes (I wrote the "Committing Changes")

Go to actions -> All workflows



Click on "Committing Changes"

github.com/hafsalman/Git-action-Repo/actions/runs/18613808781

hafsalman / Git-action-Repo

Code Issues Pull requests Actions Projects Wiki Security Insights Settings

Deploy Static Website

Committing changes #1

Summary

Jobs

deploy

Run details

Usage

Workflow file

Triggered via push 1 minute ago

hafsalman pushed · 1195700 main

Status Success

Total duration 18s

Artifacts 1

main.yml

on: push

deploy 10s

https://hafsalman.github.io/Git-action-Rep...

Artifacts

Produced during runtime

Name	Size	Digest
github-pages	879 Bytes	sha256:9504d70ac780b750a251c41ae7c87b44d436d2140e205a995a8...

Click Deploy

github.com/hafsalman/Git-action-Repo/actions/runs/18613808781/job/53075615922

hafsalman / Git-action-Repo

Code Issues Pull requests Actions Projects Wiki Security Insights Settings

Deploy Static Website

Committing changes #1

Summary

Jobs

deploy

Run details

Usage

Workflow file

deploy

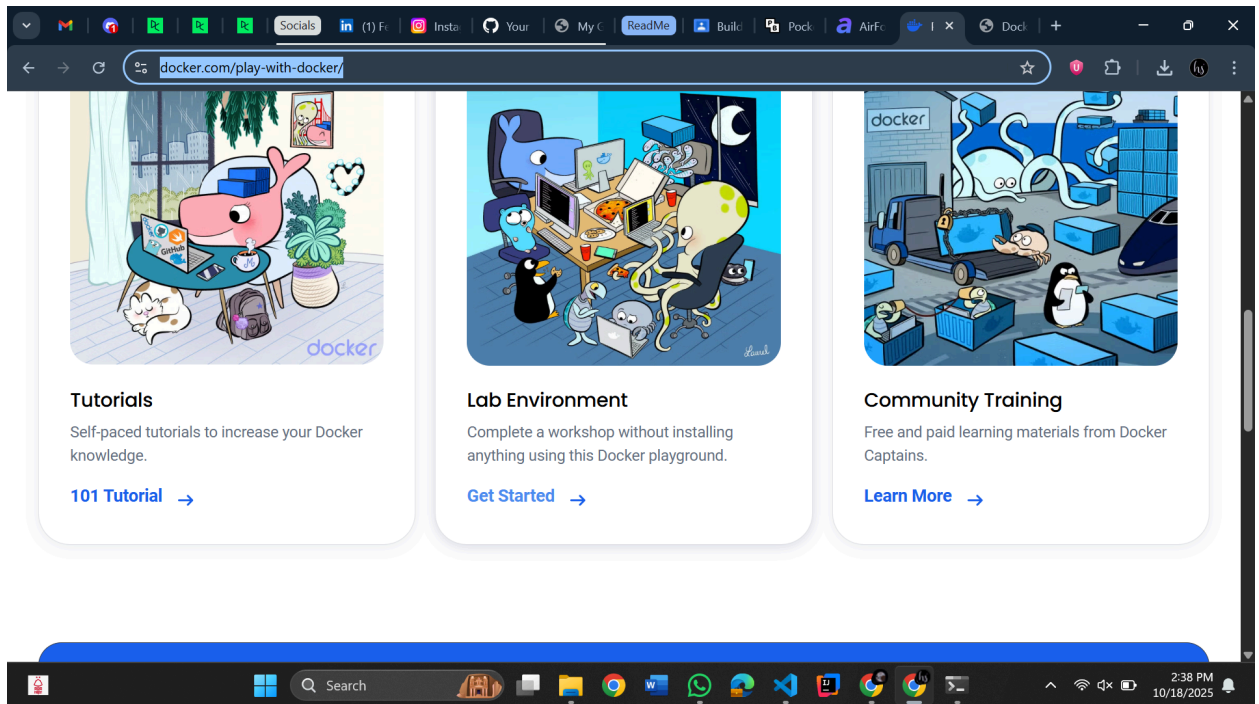
succeeded 1 minute ago in 10s

Search logs

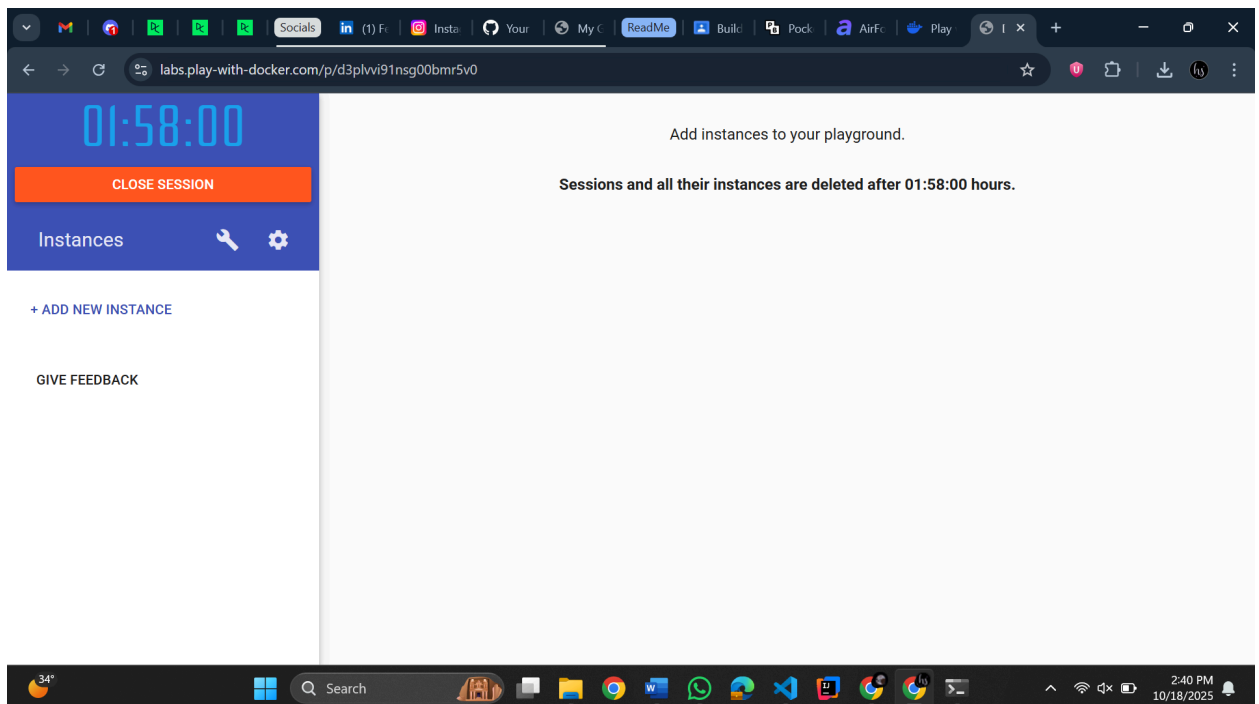
Set up job 1s

- 1 Current runner version: '2.329.0'
- 2 ▶ Runner Image Provisioner
- 7 ▶ Operating System
- 11 ▶ Runner Image
- 16 ▶ GITHUB_TOKEN Permissions
- 20 Secret source: Actions
- 21 Prepare workflow directory
- 22 Prepare all required actions
- 23 Getting action download info
- 24 Download action repository 'actions/checkout@v4' (SHA:88eba0027e820071cde6df949e0be95b4906955)

-
- The screenshot shows a web browser window with the address bar displaying 'hafs Salman.github.io/Git-action-Repo/'. The page content includes a large blue heading 'Welcome to My Website!', a message 'This site is hosted automatically using GitHub Actions' with a small GitHub Actions logo, and a blue button labeled 'Click Me!'.



3. Get started
4. Sign up
5. Give username
6. Start



You will end up here

7. Open the docker word file (I will upload that on the repo as well)
8. Create instance

01:56:18

CLOSE SESSION

Instances



+ ADD NEW INSTANCE

GIVE FEEDBACK

Sessions and

labs.play-with-docker.com/p/d3plvvi91nsg00bmr5v0#d3plvvi9_d3pm1p291nsg00bmr680

01:55:55
CLOSE SESSION

Instances
+ ADD NEW INSTANCE

192.168.0.29
node1

GIVE FEEDBACK

d3plvvi9_d3pm1p291nsg00bmr680

IP
192.168.0.29
OPEN PORT

Memory
SSH
ssh ip172-18-0-28-d3plvvi91nsg00bmr5v0@direct.labs.play

DELETE EDITOR

```
#####
# WARNING!!!!
# This is a sandbox environment. Using personal credentials
# is HIGHLY! discouraged. Any consequences of doing so are
# completely the user's responsibilities.
#
# The PWD team.
#####
[node1] (local) root@192.168.0.29 ~
$ docker --version
Docker version 27.3.1, build ce12230
[node1] (local) root@192.168.0.29 ~
$
```

34° Search 2:42 PM 10/18/2025

Run docker --version

labs.play-with-docker.com/p/d3plvvi91nsg00bmr5v0#d3plvvi9_d3pm1p291nsg00bmr680

01:54:45
CLOSE SESSION

Instances
+ ADD NEW INSTANCE

192.168.0.29
node1

GIVE FEEDBACK

d3plvvi9_d3pm1p291nsg00bmr680

IP
192.168.0.29
OPEN PORT

Memory
SSH
ssh ip172-18-0-28-d3plvvi91nsg00bmr5v0@direct.labs.play

DELETE EDITOR

```
#####
[node1] (local) root@192.168.0.29 ~
$ docker --version
Docker version 27.3.1, build ce12230
[node1] (local) root@192.168.0.29 ~
$ docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
4b3ffd8ccb52: Pull complete
Digest: sha256:66460d557b25769b102175144d538d88219c077c678a49af4afca6fbfclb5252
Status: Downloaded newer image for ubuntu:latest
docker.io/library/ubuntu:latest
[node1] (local) root@192.168.0.29 ~
$
```

34° Search 2:43 PM 10/18/2025

01:54:09

CLOSE SESSION

Instances

+ ADD NEW INSTANCE

192.168.0.29
node1

GIVE FEEDBACK

d3plvvi9_d3pm1p291nsg00bmr680

IP
192.168.0.29

OPEN PORT

Memory

CPU

SSH
ssh ip172-18-0-28-d3plvvi91nsg00bmr5v0@direct.labs.play

DELETE

EDITOR

```
[node1] (local) root@192.168.0.29 ~
$ docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
4b3ffd80cb52: Pull complete
Digest: sha256:66460d557b25769b102175144d538d88219c077c678a49af4afca6fbfc1b5252
Status: Downloaded newer image for ubuntu:latest
docker.io/library/ubuntu:latest
[node1] (local) root@192.168.0.29 ~
$ docker images
REPOSITORY TAG IMAGE ID CREATED SIZE
ubuntu latest 97bed23a3497 2 weeks ago 78.1MB
[node1] (local) root@192.168.0.29 ~
$
```

```
[node1] (local) root@192.168.0.29 ~
$ docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
8c7716127147: Downloading 298.3kB/29.78MB
250b90fb2b9a: Downloading 10.07MB/32.9MB
5d8ea9f4c626: Download complete
58d144c4badd: Waiting
b459da543435: Waiting
8da8ed3552af: Waiting
54e822d8ee0c: Waiting
$
```

```
58d144c4badd: Pull complete
b459da543435: Pull complete
8da8ed3552af: Pull complete
54e822d8ee0c: Pull complete
Digest: sha256:3b7732505933ca591ce4a6d860cb713ad96a3176b82f7979a8dfa9973486a0d6
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
[node1] (local) root@192.168.0.29 ~
$ docker images
REPOSITORY TAG IMAGE ID CREATED SIZE
nginx latest 07ccdb783875 10 days ago 160MB
ubuntu latest 97bed23a3497 2 weeks ago 78.1MB
[node1] (local) root@192.168.0.29 ~
$
```

```
Search
```

```
54e822d8ee0c: Pull complete
Digest: sha256:3b7732505933ca591ce4a6d860cb713ad96a3176b82f7979a8dfa9973486a0d6
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
[node1] (local) root@192.168.0.29 ~
$ docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
nginx         latest   07ccdb783875   10 days ago    160MB
ubuntu        latest   97bed23a3497   2 weeks ago    78.1MB
[node1] (local) root@192.168.0.29 ~
$ docker create --name myubuntu ubuntu
eabb8709cddec6057fd9e095085612d02e891a5ac7b7518145a4684d7f572d7d
[node1] (local) root@192.168.0.29 ~
$
```



```
[node1] (local) root@192.168.0.29 ~
$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS   NAMES
[node1] (local) root@192.168.0.29 ~
$
```

```
[node1] (local) root@192.168.0.29 ~
$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS   NAMES
[node1] (local) root@192.168.0.29 ~
$ docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED        STATUS      PORTS   NAMES
eabb8709cdde   ubuntu   "/bin/bash"   2 minutes ago   Created             myubuntu
[node1] (local) root@192.168.0.29 ~
$
```

```
[node1] (local) root@192.168.0.29 ~
$ docker run myubuntu
Unable to find image 'myubuntu:latest' locally
docker: Error response from daemon: pull access denied for myubuntu, repository does not exist or may require 'docker login': denied: requested access to the resource is denied.
See 'docker run --help'.
[node1] (local) root@192.168.0.29 ~
$
```

